

MTH 9875 The Volatility Surface: Fall 2026

Lecture 1: Stylized facts

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Outline of lecture 1

- Power laws in finance
- Zipf's law
- The 80/20 rule
- Power laws in financial data
- Financial time series: more stylized facts
- The volatility surface
 - Power-law scaling of short-dated smiles
 - The skew-stickiness ratio
 - The volatility envelope

Power laws

S has a power-law distribution (or distribution with a power-law tail) if for large x ,

$$\Pr(S > x) \sim \frac{1}{x^\zeta}$$

for some exponent ζ .

- Power-laws are ubiquitous in social sciences, and in particular in finance.
- See https://en.wikipedia.org/wiki/Power_law.
- See also the excellent articles by Newman^[12] and Visser^[13].

Zipf's Law

"The frequency of any word is inversely proportional to its rank in the frequency table."

$$\Pr(S > x) \sim \frac{1}{x^\zeta} \quad \text{with } \zeta \approx 1$$

This empirical law applies to the sizes of cities and many other economic variables. Here are some others:

- Amazon book rank
- Firm sizes
- Income and wealth
- CEO compensation.

Aggregation properties

If X_i are independent power-law distributed random variables,

$$\zeta_{X_1 + \dots + X_n} = \min(\zeta_{X_1}, \dots, \zeta_{X_n}) \quad (1)$$

$$\zeta_{X_1 \times \dots \times X_n} = \min(\zeta_{X_1}, \dots, \zeta_{X_n}) \quad (2)$$

$$\zeta_{\max(X_1, \dots, X_n)} = \min(\zeta_{X_1}, \dots, \zeta_{X_n}) \quad (3)$$

$$\zeta_{\min(X_1, \dots, X_n)} = \zeta_{X_1} + \dots + \zeta_{X_n} \quad (4)$$

$$\zeta_{\alpha X} = \zeta_X \quad (5)$$

$$\zeta_{X^\alpha} = \frac{\zeta_X}{\alpha} \quad (6)$$

- Thus, combinations of power-law distributed random variables give rise to more power-laws.

The Pareto Principle (the 80/20 rule)

It's an often stated management principle that in many cases, roughly 80% of the effects come from 20% of the causes. Pareto originally noted that 80% of Italy's land was owned by 20% of the population. We can give many examples:

- 80% of sales come from 20% of clients.
- The top 20% of earners pay roughly 80% of Federal income taxes.
- Fixing the top 20% most-reported programming bugs eliminates 80% of errors.

The 80/20 rule reflects a power-law distribution of the underlying variable.

- Assume a density of the form $p(x) = C x^{-(\alpha+1)}$, $x > x_{\min}$ (the Pareto density).
 - Note that a *tail exponent* of α for the distribution gives an exponent $\alpha + 1$ for the density.

- Assume a minimum size of $x_{\min}$ of the random variable x .
- Then the proportion greater than x^* is given by

$$P := \Pr(x > x^*) = \frac{\int_{x^*}^{\infty} p(x) dx}{\int_{x_{\min}}^{\infty} p(x) dx} = \left(\frac{x^*}{x_{\min}} \right)^{-\alpha}.$$

- In terms of wealth W , the proportion of wealth held by this proportion of the population is given by

$$W := W(x^*) = \frac{\int_{x^*}^{\infty} x p(x) dx}{\int_{x_{\min}}^{\infty} x p(x) dx} = \left(\frac{x^*}{x_{\min}} \right)^{1-\alpha}.$$

- Expressing W in terms of P ,

$$W = P^{\frac{1-\alpha}{-\alpha}} = P^{1-\frac{1}{\alpha}}.$$

- Thus, 20% of the population controlling 80% of wealth corresponds to

$$0.8 = 0.2^{1-\frac{1}{\alpha}}$$

```
In [1]: (R <- log(.8)/log(.2))
```

```
0.138646883853214
```

```
In [2]: (alpha <- 1/(1-R))
```

```
1.16096404744368
```

We see that the 80/20 rule corresponds to a power-law with tail exponent of roughly one.

Power laws in daily SPX returns

Using R and the quantmod package, we analyze log-returns of SPX since 1927.

```
In [3]: library(repr)
options(repr.plot.height=8, repr.plot.width=13, repr.res=200)
```

```
In [4]: library(quantmod)
library(boot)
library(stinepack)
```

Loading required package: xts

Loading required package: zoo

Attaching package: 'zoo'

The following objects are masked from 'package:base':

as.Date, as.Date.numeric

Loading required package: TTR

Registered S3 method overwritten by 'quantmod':

method from
as.zoo.data.frame zoo

```
In [5]: #Get SPX and VIX data from Yahoo!
options("getSymbols.warning4.0"=FALSE)

getSymbols("^GSPC", from="1927-01-01") #Creates the time series object GSPC
```

'GSPC'

@d SPX

```
In [6]: getSymbols("^VIX", from="1927-01-01") #Creates the time series object VIX

mm <- specifyModel(Cl(GSPC)~Cl(VIX))
spxVixData <- modelData(mm) #quantmod function automatically aligns data from

vix <- spxVixData[, "Cl.VIX"]
spx <- spxVixData[, "Cl.GSPC"]

print(head(spxVixData))
print(tail(spxVixData))
```

Warning message:

"^VIX contains missing values. Some functions will not work if objects contain missing values in the middle of the series. Consider using na.omit(), na.approx(), na.fill(), etc to remove or replace them."

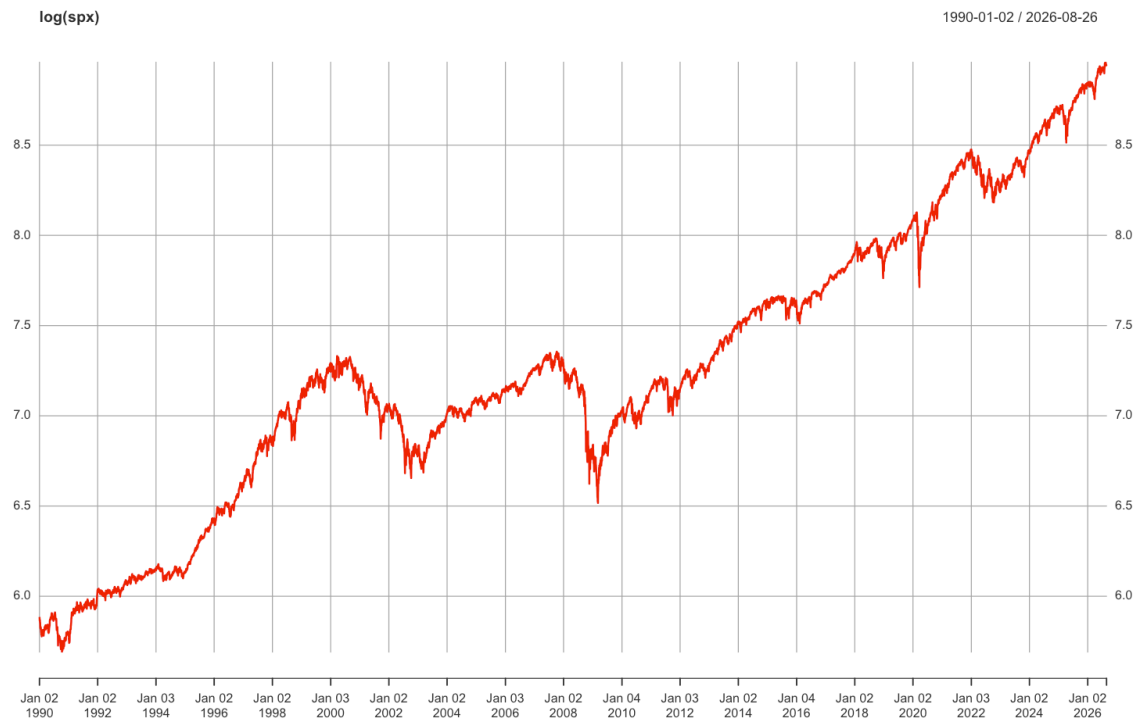
'VIX'

	CL.GSPC	CL.VIX
1990-01-02	359.69	17.24
1990-01-03	358.76	18.19
1990-01-04	355.67	19.22
1990-01-05	352.20	20.11
1990-01-08	353.79	20.26
1990-01-09	349.62	22.20
	CL.GSPC	CL.VIX
2026-08-19	7707.98	14.89
2026-08-20	7641.16	16.01
2026-08-21	7674.37	15.13
2026-08-24	7652.86	15.85
2026-08-25	7677.28	15.45
2026-08-26	7675.70	15.21

```
In [7]: plot(spx,col="royalblue")
```



```
In [8]: plot(log(spx),col="red2")
```



```
In [9]: save(spxVixData,file="spxVix.rData")
```

Histogram of SPX daily log-returns

```
In [10]: ret.spx <- log(Cl(GSPC)/lag(Cl(GSPC)))
ret.spx <- ret.spx[!is.na(ret.spx)] # Remove missing values
ret.spx <- ret.spx - mean(ret.spx)
```

```
In [11]: print(c(max(ret.spx),min(ret.spx)))
```

```
[1]  0.1534161 -0.2292424
```

```
In [12]: breaks <- seq(-.235,.155,.002)
hist.spx <- hist(ret.spx,breaks=breaks,freq=F)
```

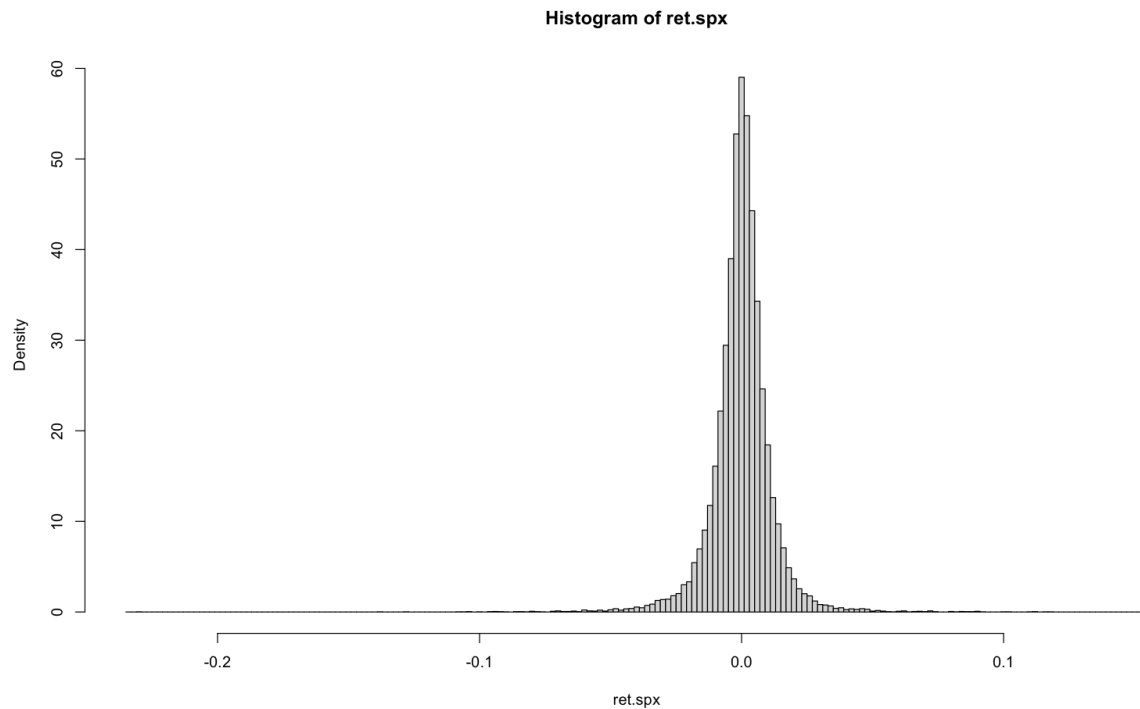


Figure 1: Histogram of SPX daily log-returns. Note the outlier in the left tail!

The cubic law of returns

```
In [13]: print(names(hist.spx))
spx.density <- hist.spx$density
mids <- hist.spx$mids
x <- abs(mids)[abs(mids)>0.005]
y <- spx.density[abs(mids)>0.005]
```

```
[1] "breaks" "counts" "density" "mids" "xname" "equidist"
```

```
In [14]: # Log-log plot
plot(log(x), log(y), ylab="log density", xlab="log abs. returns", pch=20, col="rc
```

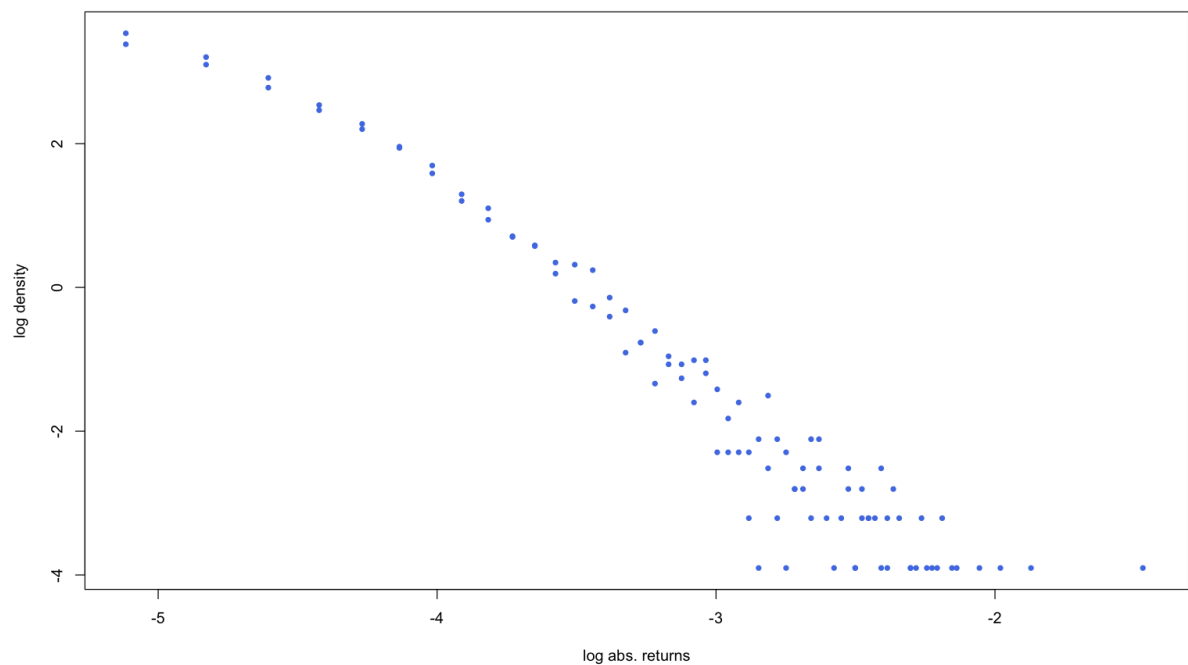


Figure 2: Log-log plot of the empirical density of absolute daily returns of SPX.

```
In [15]: # Look only at tails
cutoff <- 0.02
rh <- (mids > cutoff)
lh <- (mids < (-cutoff))

xrh <- mids[rh]
xlh <- -mids[lh]
yrh <- spx.density[rh]
ylh <- spx.density[lh]

lnx.rh <- log(xrh)[yrh>0]
lny.rh <- log(yrh)[yrh>0]
lnx.lh <- log(xlh)[ylh>0]
lny.lh <- log(ylh)[ylh>0]

rh.lm <- lm(lny.rh~lnx.rh)
lh.lm <- lm(lny.lh~lnx.lh)
```

```
In [16]: summary(rh.lm)
```

Call:

```
lm(formula = lny.rh ~ lnx.rh)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-1.7548	-0.2672	0.1127	0.3039	1.0431

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-10.3064	0.5079	-20.29	<2e-16 ***
lnx.rh	-2.8652	0.1754	-16.34	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5721 on 40 degrees of freedom

Multiple R-squared: 0.8697, Adjusted R-squared: 0.8664

F-statistic: 267 on 1 and 40 DF, p-value: < 2.2e-16

```
In [17]: plot(lnx.rh, lny.rh, ylab="log density", col="royalblue", xlab="log abs. returns",
points(lnx.lh, lny.lh, col="red2", pch=20)
abline(rh.lm, col="green4", lwd=2)
abline(lh.lm, col="pink2", lwd=2)
```

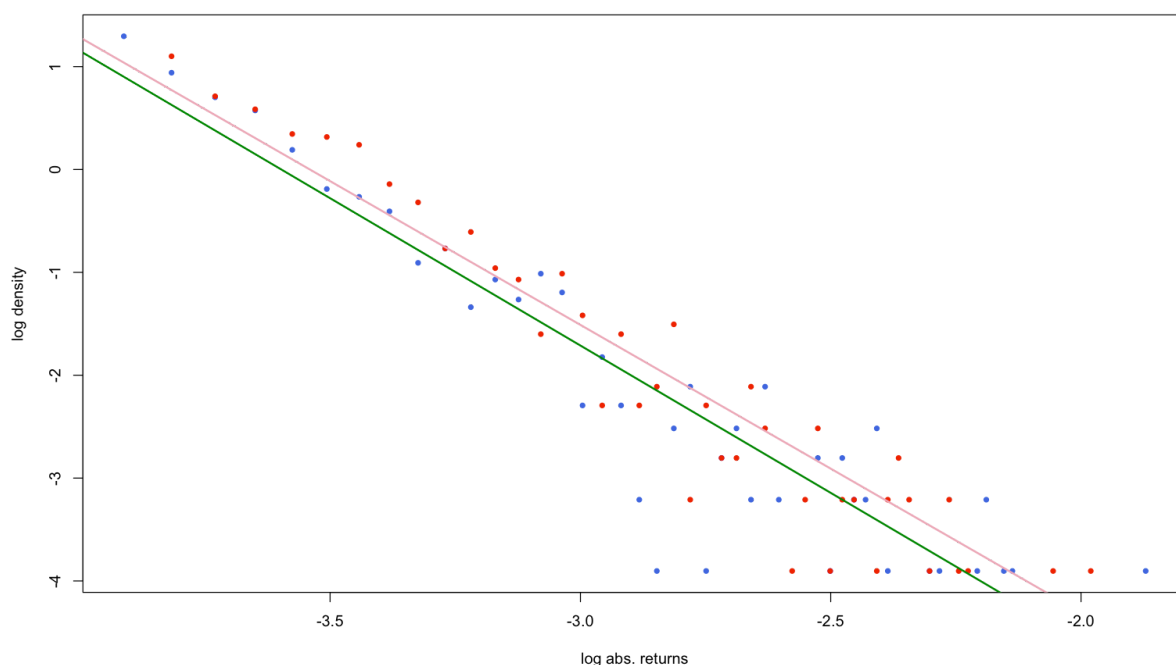


Figure 3: Log-log plot of the empirical density of absolute daily returns of SPX with a lower cutoff.

```
In [18]: # Repeat with upper cutoff
upperCutoff <- 0.08

rh <- (mids > cutoff) & (mids < upperCutoff)
lh <- (mids < (-cutoff)) & (mids > (-upperCutoff))

xrh <- mids[rh]
```

```

xlh <- -mids[lh]
yrh <- spx.density[rh]
ylh <- spx.density[lh]

lnx.rh <- log(xrh)[yrh>0]
lny.rh <- log(yrh)[yrh>0]
lnx.lh <- log(xlh)[ylh>0]
lny.lh <- log(ylh)[ylh>0]

rh.lm <- lm(lny.rh~lnx.rh)
lh.lm <- lm(lny.lh~lnx.lh)

```

```

In [19]: plot(lnx.rh, lny.rh, ylab="log density", col="royalblue", xlab="log abs. returns",
points(lnx.lh, lny.lh, col="red2", pch=20)
abline(rh.lm, col="green4", lwd=2)
abline(lh.lm, col="pink2", lwd=2)

```

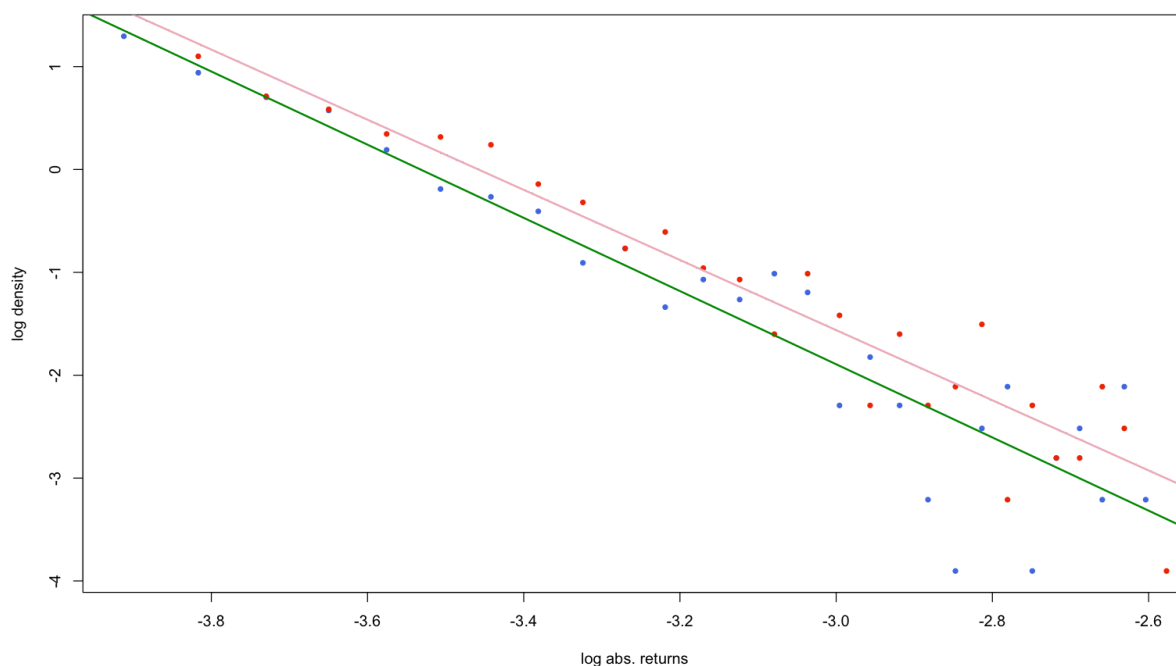


Figure 4: Log-log plot of the empirical density of SPX returns with lower and upper cutoff. The red points are from the left tail, the blue points from the right tail, the green and pink lines are linear fits. The fit slopes (rh:3.55, lh: 3.44) are consistent with a value of 3.

```

In [20]: summary(rh.lm)

```

```
Call:
lm(formula = lny.rh ~ lnx.rh)

Residuals:
    Min       1Q   Median       3Q      Max
-1.46674 -0.10620  0.04739  0.19643  1.09445

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -12.5686     0.8235  -15.26 1.71e-14 ***
lnx.rh       -3.5585     0.2617  -13.60 2.49e-13 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.5236 on 26 degrees of freedom
Multiple R-squared:  0.8768,    Adjusted R-squared:  0.872
F-statistic: 185 on 1 and 26 DF,  p-value: 2.493e-13
```

```
In [21]: summary(lh.lm)
```

```
Call:
lm(formula = lny.lh ~ lnx.lh)

Residuals:
    Min       1Q   Median       3Q      Max
-0.90030 -0.17989  0.02311  0.20816  0.69265

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -11.7902     0.6118  -19.27 < 2e-16 ***
lnx.lh       -3.4095     0.1988  -17.15 4.8e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3957 on 27 degrees of freedom
Multiple R-squared:  0.9159,    Adjusted R-squared:  0.9128
F-statistic: 294.2 on 1 and 27 DF,  p-value: 4.804e-16
```

The Student-t distribution

$$p(x) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\Gamma\left(\frac{\nu}{2}\right)} \frac{1}{\left(1 + \frac{x^2}{\nu}\right)^{\frac{\nu+1}{2}}} \sim \frac{1}{x^{\nu+1}} \text{ as } x \rightarrow \infty$$

So the tail-exponent $\zeta = \nu$, the degrees of freedom.

Obviously, this distribution has mean zero. Its variance is

$$\int x^2 p(x) dx = \frac{\nu}{\nu - 2}$$

With $\nu = 3$, the density simplifies to

$$p(x) = \frac{2}{\sqrt{3}\pi} \frac{1}{\left(1 + \frac{x^2}{3}\right)^2}$$

SPX daily log-returns: Student-t fit

```
In [22]: plot(hist.spx,xlim=c(-.05,.05),freq=F,main=NA,xlab="Log return")
sig <- as.numeric(sd(ret.spx))
scale <- function(nu){sqrt(nu/(nu-2))}
df <- 2.7
curve(dt(x*scale(df)/sig,df)*scale(df)/sig,from=-.25,to=.25,col="red2",lwd=2)
curve(dnorm(x,mean=0,sd=sig),from=-.05,to=.05,col="royalblue",lwd=2, add=T)
```

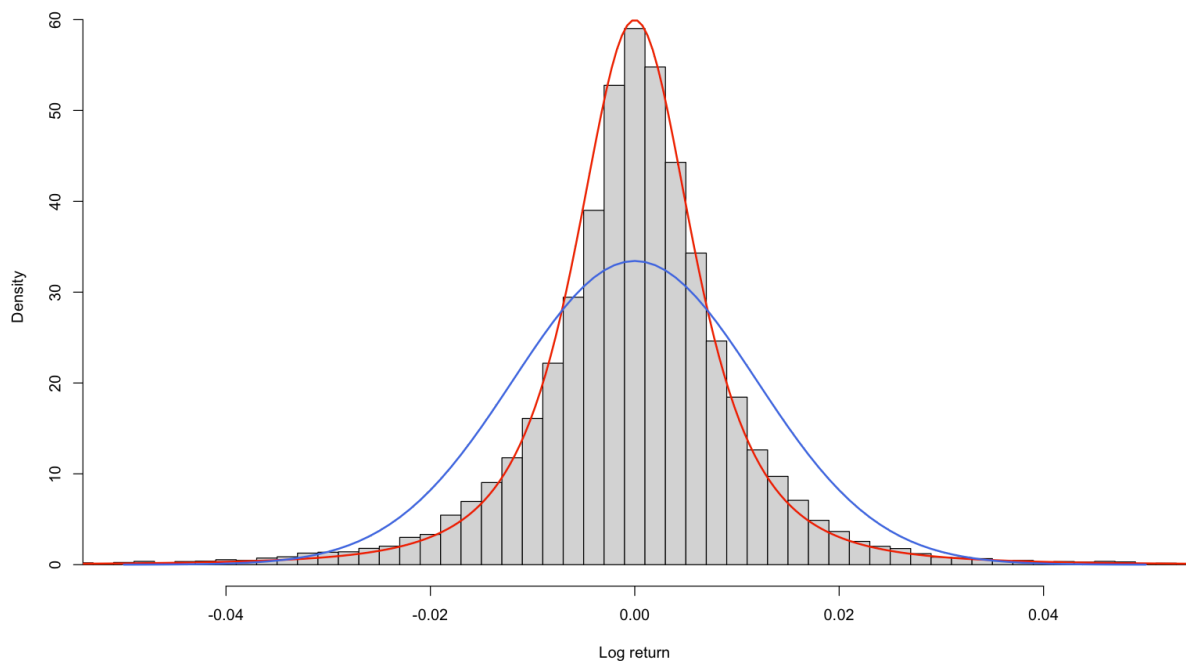


Figure 5: Student-t with $\nu = 2.7$ fits almost perfectly! Normal fit is in blue. The cubic law is confirmed.

Note on power-law tails

Although Student-t seems to fit well, the distribution of log-returns cannot have power-law tails in reality. For example, we certainly believe that the first moment of the stock price exists. That is $\mathbb{E}[S_T] = \mathbb{E}[e^x] = C$ for some $C < \infty$ where $x = \log S_T / F$. If the distribution of log-returns is power-law with tail exponent ν , then $\mathbb{E}[x^\alpha] = \infty$ if $\alpha > \nu$ and in particular $\mathbb{E}[e^x] > \infty$.

This is obviously true more generally in physics: one cannot have a true power-law in a system with finite physical extent. Power-laws describe approximate behavior up to some cutoff.

SPX weekly log-returns

```
In [23]: pxw <- Cl(to.weekly(GSPC)) # Built-in quantmod function to get close
retw <- Delt(pxw,type="log")[-1]
retw <- retw-mean(retw)
c(min(retw),max(retw))
breaks <- seq(-.215,.155,.005)
spx.histw <- hist(retw,breaks=breaks,plot=F)
sigw <- as.numeric(sd(retw))
scale <- function(nu){sqrt(nu/(nu-2))}
```

-0.206550228462218 · 0.14820443975237

```
In [24]: df <- 3.5
plot(spx.histw,xlim=c(-.215,.155),freq=F,main=NA,xlab="Weekly log return")
curve(dt(x*scale(df)/sigw,df)*scale(df)/sigw,from=-.25,to=.25,col="red2",lwd=2)
curve(dnorm(x,mean=0,sd=sigw),from=-.25,to=.25,col="royalblue", add=T,lwd=2)
```

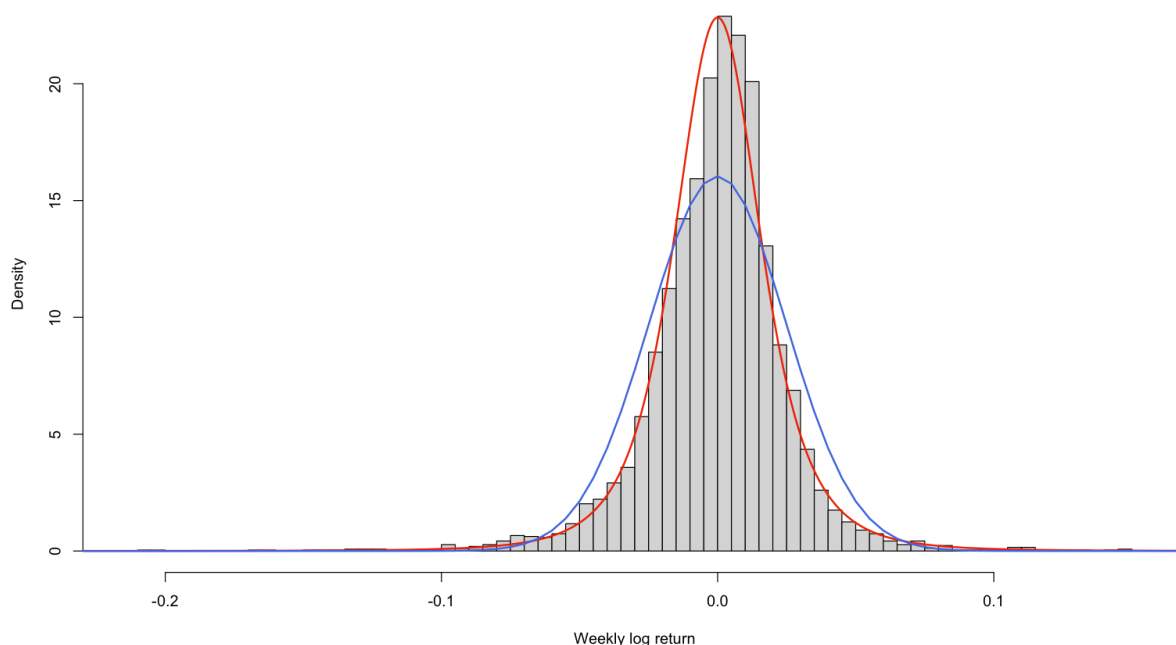


Figure 6: Student-t with $\nu = 3.5$ in red. Normal fit is in blue.

SPX monthly log-returns

```
In [25]: pxm <- Cl(to.monthly(GSPC)) # Built-in quantmod function to get close
retm <- Delt(pxm,type="log")[-1]
retm <- retm-mean(retm)
c(min(retm),max(retm))
breaks <- seq(-.365,.335,.01)
spx.histm <- hist(retm,breaks=breaks,plot=F)
sigm <- as.numeric(sd(retm))
scale <- function(nu){sqrt(nu/(nu-2))}
```

-0.36098118327638 · 0.325163015349159

```
In [26]: df <- 3.5
plot(spx.histm,xlim=c(-.365,.335),freq=F,main=NA,xlab="Monthly log return")
curve(dt(x*scale(df)/sigm,df)*scale(df)/sigm,from=-.35,to=.35,col="red2", add=T)
curve(dnorm(x,mean=0,sd=sigm),from=-.35,to=.35,col="royalblue", add=T,lwd=2)
```

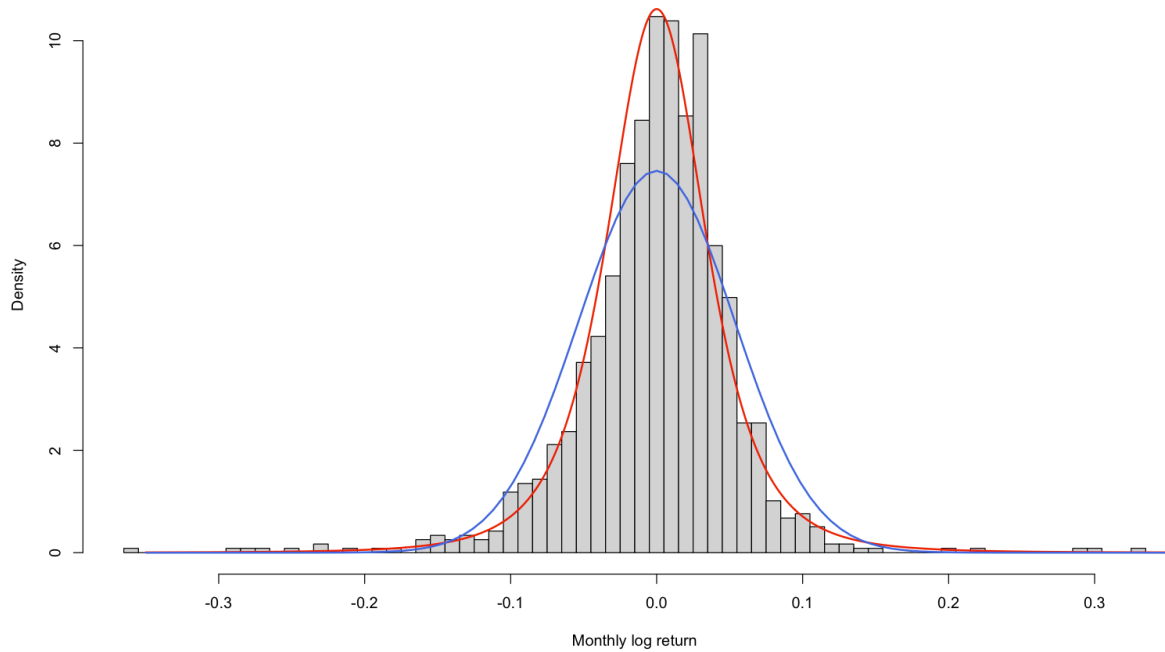


Figure 7: Student-t with $\nu = 3.2$ in red. Normal fit is in blue.

A digression: the Fokker-Planck equation

The Fokker-Planck equation describes the evolution of the probability density in a diffusion process. Specifically, suppose

$$dX_t = \mu(X_t, t) dt + \sigma(X_t, t) dW_t$$

Then, applying Itô's Formula,

$$df = \frac{\partial f}{\partial t} dt + \mu(X_t, t) \frac{\partial f}{\partial X_t} dt + \frac{\sigma(X_t, t)^2}{2} \frac{\partial^2 f}{\partial X_t^2} dt + \sigma(X_t, t) \frac{\partial f}{\partial X_t} dW_t$$

We now choose a special $f(\cdot)$. Let $f(z, t) = \delta(z - x)$ (independent of t). Then

$$\mathbb{E}[f(X_t, t)] = \int_{-\infty}^{+\infty} dz p(z, t) \delta(z - x) = p(x, t)$$

Taking expectations of Itô's Lemma, noting that $\partial_t f(z, t) = 0$, we obtain (if the density decays suitably fast at $\pm\infty$),

$$\begin{aligned} & \frac{d\mathbb{E}[f(X_t, t)]}{dt} \\ &= \mathbb{E} \left[\mu(z, t) \frac{\partial f(z, t)}{\partial z} + \frac{\sigma^2(z, t)}{2} \frac{\partial^2 f(z, t)}{\partial z^2} \right] \\ &= \int_{-\infty}^{+\infty} dz p(z, t) \left\{ \mu(z, t) \frac{\partial f(z, t)}{\partial z} + \frac{\sigma^2(z, t)}{2} \frac{\partial^2 f(z, t)}{\partial z^2} \right\} \\ &= \int_{-\infty}^{+\infty} dz f(z, t) \left\{ -\frac{\partial}{\partial z} (\mu(z, t) p(z, t)) + \frac{1}{2} \frac{\partial^2}{\partial z^2} (\sigma^2(z, t) p(z, t)) \right\} \end{aligned}$$

Substituting $f(z, t) = \delta(z - x)$ gives

The forward Kolmogorov or Fokker-Planck equation

$$\frac{dp(x, t)}{dt} = -\frac{\partial}{\partial x} (\mu(x, t) p(x, t)) + \frac{1}{2} \frac{\partial^2}{\partial x^2} (\sigma^2(x, t) p(x, t))$$

Where do power-laws come from?

The following robust heuristic derivation of Zipf's Law is due to Gabaix^[8]. Assume

(1)

$$dX_t = \mu X_t dt + \nu X_t dZ_t$$

where X_t is size.

Let $f(x, t)$ be the density (of sizes for example). In steady state, $\partial_t f(\cdot) = 0$ so put $f(x, t) = f(x)$ and the forward Kolmogorov (Fokker-Planck) equation then reads:

$$-\partial_x (\mu x f(x)) + \partial_{x,x} \left(\frac{\nu^2}{2} x^2 f(x) \right) = 0$$

Try

$$f(x) = \frac{C}{x^{\zeta+1}}$$

Then

$$\frac{C}{x^{\zeta+1}} \left\{ \mu \zeta + \frac{\nu^2}{2} (\zeta - 1) \zeta \right\} = 0$$

Then either $\zeta = 0$ or

$$\zeta = 1 - \frac{2\mu}{\nu^2}.$$

The choice $\zeta = 0$ gives a logarithmic function that diverges at $x \rightarrow 0$ and as $x \rightarrow \infty$.

The choice $\zeta = 1 - \frac{2\mu}{\nu^2}$ gives a density that diverges as $x \rightarrow 0$, so cut off at S_{min} .

$$\int f(x) dx = \int_{S_{min}}^{\infty} \frac{C}{x^{\zeta+1}} dx = \frac{C}{\zeta S_{min}^{\zeta}} = 1$$

Then we obtain the power law

$$f(x) = \zeta S_{min}^{\zeta} \frac{1}{x^{\zeta+1}}; \tilde{F}(x) = \left(\frac{S_{min}}{x} \right)^{\zeta}.$$

where $\tilde{F}$ is the tail distribution. Now, to estimate ζ , compute the average size

$$\bar{S} = \int x f(x) dx = \frac{\zeta}{\zeta - 1} S_{min}.$$

Then

$$\zeta = \frac{1}{1 - \frac{S_{min}}{\bar{S}}}$$

which gives us our result if $S_{min} \ll \bar{S}$.

The Gabaix result in words

If the average size of something is much bigger than the minimum size, and assuming the growth process (1), the distribution of sizes is power-law with tail exponent approximately 1.

Ané and Geman

According to Ané and Geman^[1], returns are normal when measured in business time.

- Their procedure has been shown to be defective and their conclusion is widely considered to be false.
- Gillemot, Farmer, and Lillo^[11] is one paper that shows that the methodology of Ané and Geman is incorrect.
- Nevertheless, returns normalized by integrated variance are very close to normal as we will see below.

The Oxford-Man dataset

- The Oxford-Man Institute of Quantitative Finance used to make historical realized variance (RV) estimates freely available
 - These estimates were updated daily.
 - Each day, for 31 different indices, all trades and quotes were used to estimate realized (or integrated) variance over the trading day from open to close.
 - The opening and closing prices were also recorded.
- Using this dataset, we may investigate the distribution of intraday returns normalized by integrated variance empirically.

Tail distribution of normalized returns

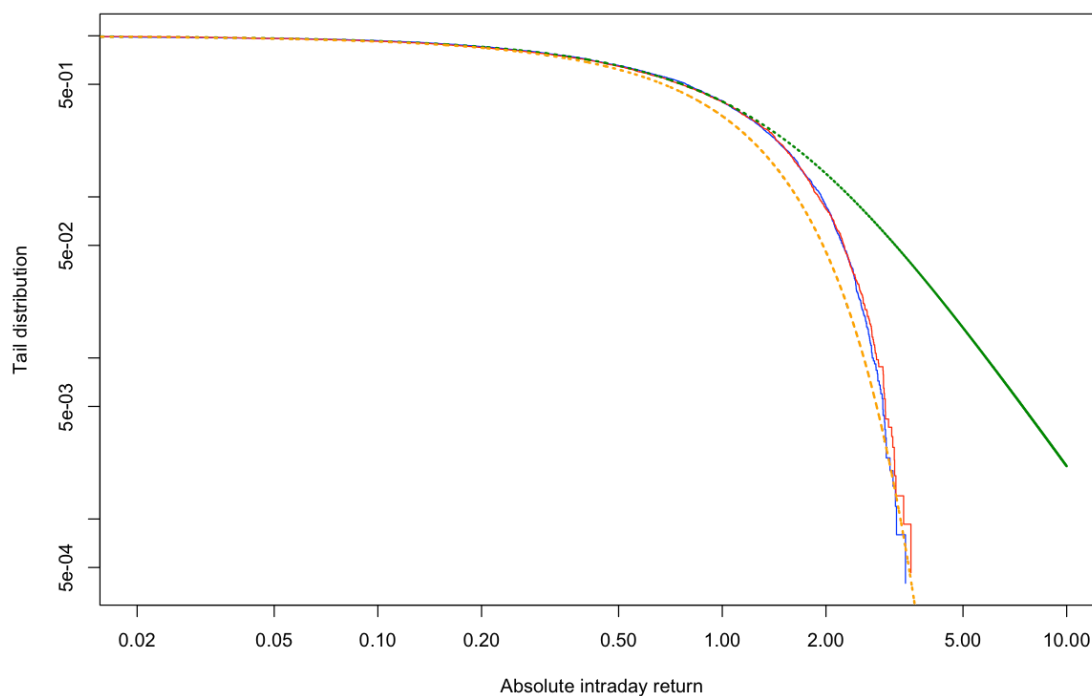


Figure 8: The blue line is the empirical distribution of normalized intraday returns from Jan-2000 to Dec-2009; the red line from Jan-2010 to today. The orange line is an $N(0, 1)$ fit and the green line Student-t with 3 degrees of freedom, consistent with the cubic law of returns. We see that the tail distribution seems to be stable over time and quite consistent with $N(0, 1)$.

Rationale for stochastic volatility

- Though the conclusion of Ané and Geman may be (strictly speaking) incorrect, defining business time instead as proportional to integrated variance gives results

that support their claim.

- It is then also very natural to model the evolution of the underlying as

$$\frac{dS_t}{S_t} = \sigma_t dZ_t$$

where σ_t is some stochastic process - in other words, stochastic volatility.

- Itô's Formula gives

$$d \log S_t = \sigma_t dZ_t + \frac{1}{2} \sigma_t^2 dt.$$

- By Itô's isometry, the quadratic variation of $\log S$ is given by

$$d\langle \log S \rangle_t = \sigma_t^2 dt =: V_t dt.$$

- V_t is termed *instantaneous variance*.
- So stochastic volatility models may be written in general as

$$\frac{dS_t}{S_t} = \sqrt{V_t} dZ_t,$$

with $V_t = \partial_t \langle \log S \rangle_t$.

Power laws in high-frequency returns

It is claimed that high-frequency returns obey the cubic law, consistent with daily returns.

For a good introduction to the analysis of empirical financial data, see Bouchaud and Potters^[3].

- We analyze 16 days of CSCO trade data from 01-Aug-2018 to 22-Aug-2018
- over 350,000 trades.
- We compute price changes over intervals of a given number of trades.

Price changes after 1,000 trades



Figure 9: A histogram of CSCO price changes after 1,000 trades. The red line is a fit of the normal density.

- We see from Figure 9 that the empirical distribution of high-frequency returns in trading time has a high peak and fat tails.
- Contrast this with the claim of Ané and Geman that returns in trading time are normal.
- In Figure 10 below, we see that returns in trading time have fat tails.

Log-log plot of CSCO prices changes

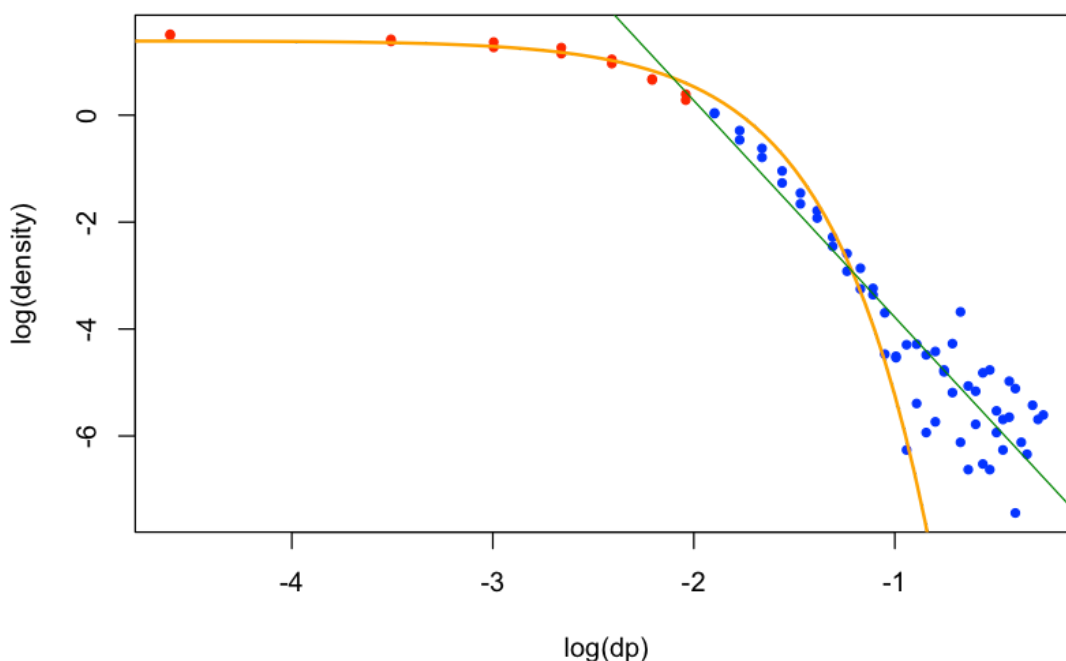


Figure 10: The green line is a (power-law) fit to the blue points; the slope is around -4.05 , completely consistent with the cubic law. The orange curve is the fitted normal density.

The Gabaix (2006) market impact explanation of the cubic law

The argument of Gabaix^[8] goes as follows:

Market impact is proportional to the square-root of trade size so

$$\frac{\Delta S}{S} \sim \sqrt{\mathcal{V}}$$

Then, the dollar cost of a trade $\sim \mathcal{V}^{3/2}$. It seems reasonable to suppose that the average dollar impact of a fund's trades should be proportional to the size X of the fund. So

$$\mathcal{V}^{3/2} \sim X; \mathcal{V} \sim X^{2/3}$$

Fund sizes are assumed to be Zipf-distributed so $\zeta_X \approx 1$.

Applying the rule from above

$$\zeta_{X^\alpha} = \frac{\zeta_X}{\alpha},$$

gives

$$\zeta_V = \frac{\zeta_X}{2/3} = \frac{3}{2}$$

The price return r satisfies

$$r = \frac{\Delta S}{S} \sim \sqrt{V} \sim X^{1/3}$$

and applying the same rule again gives

$$\zeta_r = \frac{\zeta_X}{1/3} = 3$$

both consistent with empirical estimates.

Distribution of CSCO trade sizes

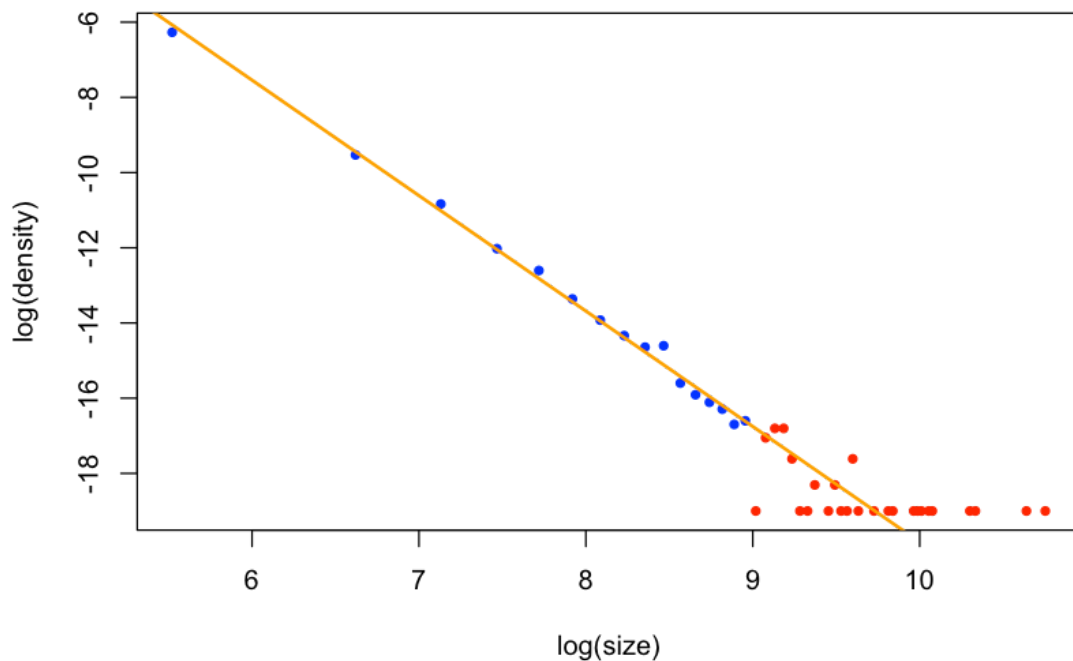


Figure 11: CSCO on 07-Sep-2006: Distribution of trade sizes. The orange line is a power-law fit to the blue points; the slope is roughly -3.07 which is not entirely inconsistent with $-5/2$.

Distribution of trade sizes for NYSE, Paris, and LSE

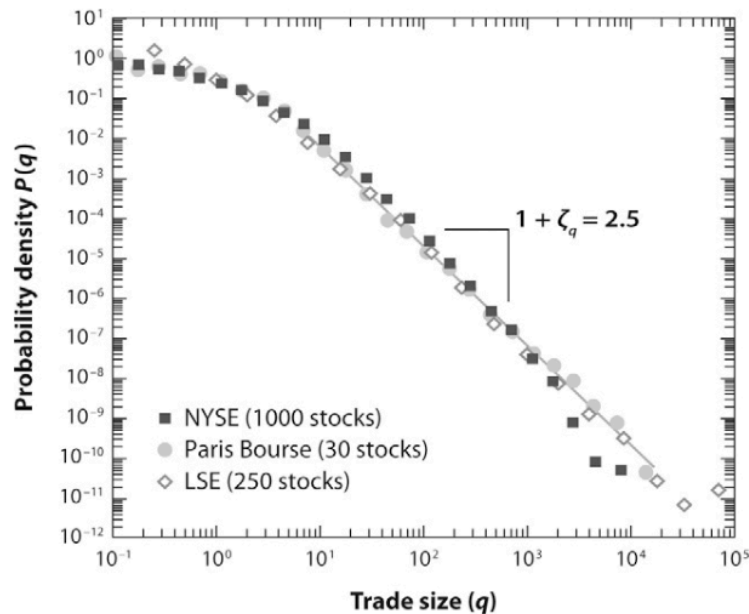


Figure 5

Probability density of normalized individual transaction sizes q for three stock markets: the New York Stock Exchange (NYSE) for 1994–1995, the London Stock Exchange (LSE) for 2001, and the Paris Bourse for 1995–1999. Ordinary-least-squares fit yields $\ln P(x) = -(1 + \zeta_q) \ln x + \text{constant}$ for $\zeta_q = 1.5 \pm 0.1$. This gives a probability density function $P(x) \sim x^{-(1+\zeta_q)}$ and a countercumulative distribution function $P(q > x) \sim x^{-\zeta_q}$. The three stock markets appear to have a common distribution of volume, with a power law exponent of 1.5 ± 0.1 . The horizontal axis shows individual volumes that are up to 10^4 times larger than the absolute deviation, $|\bar{q} - \bar{q}|$. Figure taken from Gabaix et al. 2006.

Figure 12: Distribution of market order sizes (from Gabaix^[7])

- An exponent of roughly $5/2$ in the tail of the density corresponds to a power-law tail in the CDF of approximately $3/2$, consistent with the Gabaix story.

Financial time series: stylized facts

From Cont (2001)^[5]:

the seemingly random variations of asset prices do share some quite non-trivial statistical properties. Such properties, common across a wide range of instruments, markets and time periods are called *stylized empirical facts*.

Cont (2001)^[4] lists the following stylized facts

- **Absence of autocorrelation:** (linear) autocorrelations of asset returns are often insignificant, except for very small intraday time scales.

- **Heavy tails:** the (unconditional) distribution of returns seems to display a power-law or Pareto-like tail, with a tail index which is finite, higher than two and less than five for most data sets studied. In particular this excludes stable laws with infinite variance and the normal distribution.
- **Gain/loss asymmetry:** one observes large drawdowns in stock prices and stock index values but not equally large upward movements.
- **Aggregational Gaussianity:** as one increases the time scale Δt over which returns are calculated, their distribution looks more and more like a normal distribution. In particular, the shape of the distribution is not the same at different time scales.
- **Intermittency:** returns display, at any time scale, a high degree of variability. This is quantified by the presence of irregular bursts in time series of a wide variety of volatility estimators.
- **Volatility clustering:** different measures of volatility display a positive autocorrelation over several days, which quantifies the fact that high-volatility events tend to cluster in time.
- **Conditional heavy tails:** even after correcting returns for volatility clustering (e.g. via GARCH-type models), the residual time series still exhibit heavy tails. However, the tails are less heavy than in the unconditional distribution of returns.
- **Slow decay of autocorrelation in absolute returns:** the autocorrelation function of absolute returns decays slowly as a function of the time lag, roughly as a power law with an exponent $\beta \in [0.2, 0.4]$. This is sometimes interpreted as a sign of long-range dependence.
- **Leverage effect:** most measures of volatility of an asset are negatively correlated with the returns of that asset.
- **Volatility/volume correlation:** trading volume is correlated with all measures of volatility.
- **Asymmetry in time scales:** coarse-grained measures of volatility predict fine-scale volatility better than the other way round.

To quote Rama Cont again,

these stylized facts are so constraining that it is not easy to exhibit even an *ad hoc* stochastic process which possesses the same set of properties and

one has to go to great lengths to reproduce them with a model.

Time series of SPX log-returns: Volatility clustering

```
In [27]: ret1 <- ret.spx["19980101::"] #Subset returns to include only the modern era
```

```
In [28]: plot(ret1,main=NA,col="red")
```

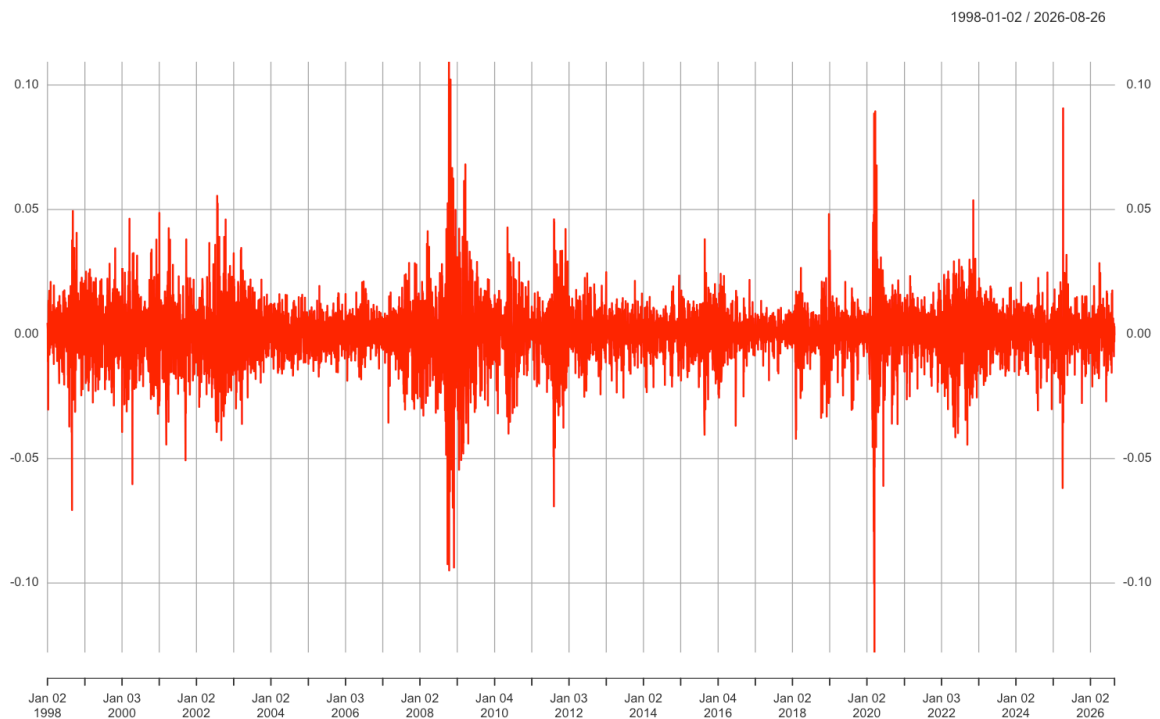


Figure 13: Daily log-returns of SPX. Note the intermittency and volatility clustering!

Autocorrelation of SPX returns

```
In [29]: acf.r <- acf(ret1,main=NA)
```

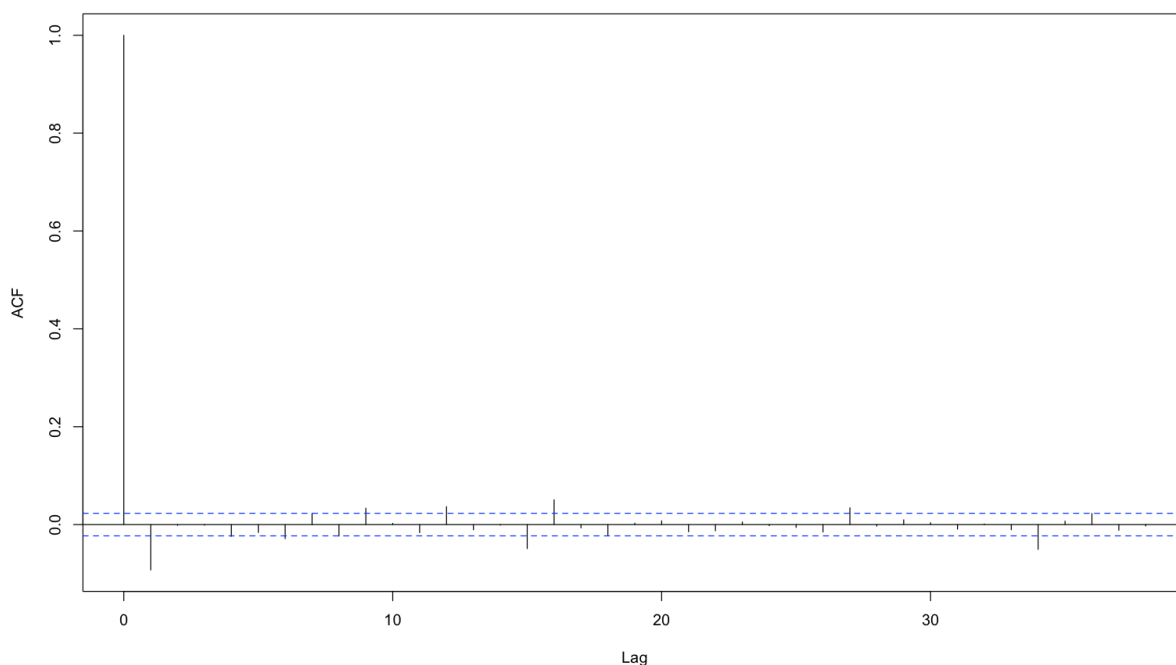


Figure 14: No significant autocorrelation in returns.

Autocorrelation of SPX absolute log-returns

It is a stylized fact that the autocorrelation function (ACF) of absolute log-returns decays as a power-law. However, we will see below that this widespread belief is probably not justified.

```
In [30]: acf.rabs <- acf(abs(ret1),main=NA,plot=F)
logacf.rabs <- log(acf.rabs$acf)[-1]
loglag.rabs <- log(acf.rabs$lag)[-1]
```

```
In [31]: plot(loglag.rabs,logacf.rabs,xlab="log(lag)",ylab="log(acf)",pch=20,col="roy",
print(acfrabs.lm <- lm(logacf.rabs[-(1:10)]~loglag.rabs[-(1:10)]))
abline(acfrabs.lm,col="red2",lwd=2)
```

Call:

```
lm(formula = logacf.rabs[-(1:10)] ~ loglag.rabs[-(1:10)])
```

Coefficients:

(Intercept)	loglag.rabs[-(1:10)]
-0.2436	-0.4435

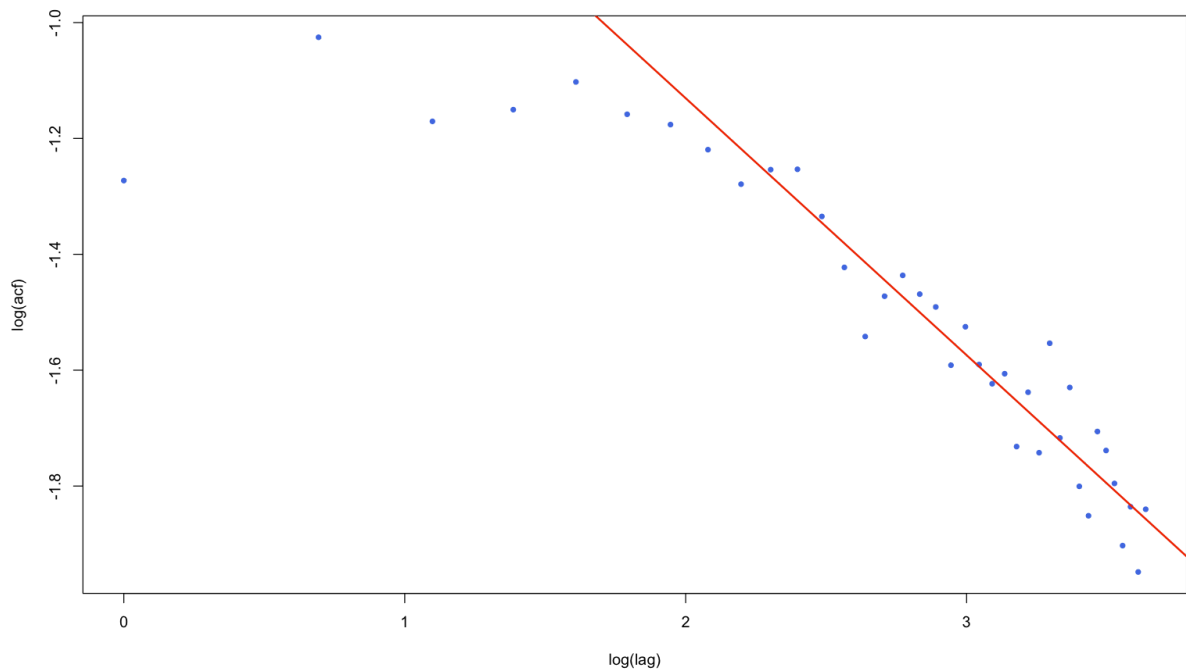


Figure 15: Slope of fit is around -0.44 .

- It is often concluded from fits like this that the autocorrelation function of volatility decays roughly as $t^{-\alpha}$ with $\alpha \approx 0.4$.

Autocorrelation of SPX squared log-returns

```
In [32]: acf.r2 <- acf(ret1^2,main=NA,plot=F)
logacf.r2 <- log(acf.r2$acf)[-1]
loglag.r2 <- log(acf.r2$lag)[-1]
plot(loglag.r2,logacf.r2,xlab="log(lag)",ylab="log(acf)",pch=20,col="royalbl1",
      (acfr2.lm <- lm(logacf.r2[-(1:10)]~loglag.r2[-(1:10)]) )
      abline(acfr2.lm,col="red2",lwd=2))
```

Call:

```
lm(formula = logacf.r2[-(1:10)] ~ loglag.r2[-(1:10)])
```

Coefficients:

(Intercept)	loglag.r2[-(1:10)]
0.2413	-0.7151

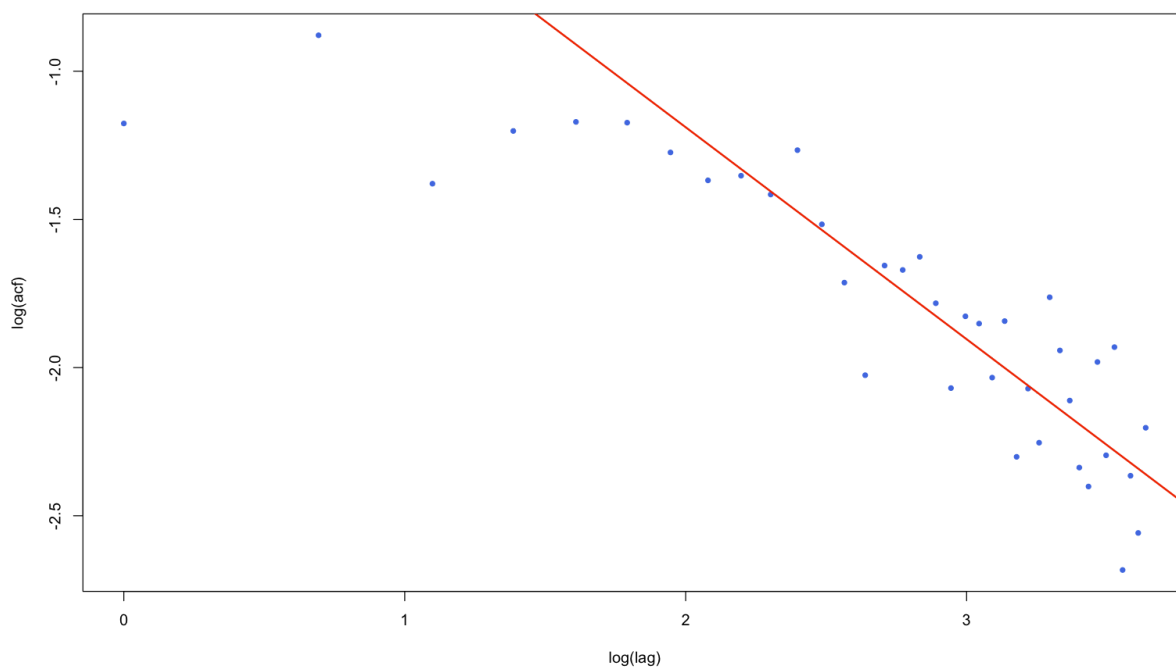


Figure 16: Slope of fit is around -0.72 so autocorrelation seems to decay roughly as $t^{-.72}$.

Decay of the volatility autocorrelation function (ACF)

That the autocorrelation function of volatility decays as a power-law is more or less established as a stylized fact. According to our work^[10] on rough volatility using such realized variance time series, the empirical ACF of volatility does not decay as a power-law.

- In fact, the log of SPX realized variance has the following amazingly simple scaling property:

$$m(q, \Delta) := \langle |\log \sigma_{t+\Delta} - \log \sigma_t|^q \rangle = A \Delta^{qH}$$

where $\langle \cdot \rangle$ denotes a sample average.

- This scaling property suggests that we model the log of realized variance as fractional Brownian motion (fBm), or more generally as fractional Ornstein-Uhlenbeck (fOU) with a very long mean reversion time.
- According to Garnier and Sølna^[9], if the log of volatility follows a fOU process, the autocorrelation function of realized variance for short times looks like

(2)

$$\rho(\Delta) \sim 1 - \frac{1}{\Gamma(1+2H)} \left(\frac{\Delta}{\epsilon} \right)^{2H}.$$

Predicted vs empirical autocorrelation function

- Squared returns can be regarded as just very noisy estimates of integrated variance.
 - We believe the characteristic timescale ϵ should be very long.
- It therefore makes sense to superimpose the short-dated approximation (2) of $\rho(\Delta)$ on the empirical ACF of squared returns.

We perform the regression:

$$\log(1 - \rho(\Delta)) \sim \log \Delta.$$

```
In [33]: y <- log(1-acf.rabs$acf)
x <- log(acf.rabs$lag)

(fit.lm <- lm(y[-(1:2)]~x[-(1:2)]))
a <- fit.lm$coef[1]
b <- fit.lm$coef[2]
```

Call:

```
lm(formula = y[-(1:2)] ~ x[-(1:2)])
```

Coefficients:

```
(Intercept)    x[-(1:2)]
   -0.5374         0.1003
```

Here is the result:

```
In [34]: plot(loglag.rabs, logacf.rabs, xlab="log(lag)", ylab="log(acf)", pch=20, col="roy",
             xlim=c(0,4.5), ylim=c(-1.9,-.8))
abline(acfrabs.lm, col="red2", lwd=2)
curve(log(1-exp(a+b*x)), from=0, to=4.5, add=T, col="green4", lwd=2)
```

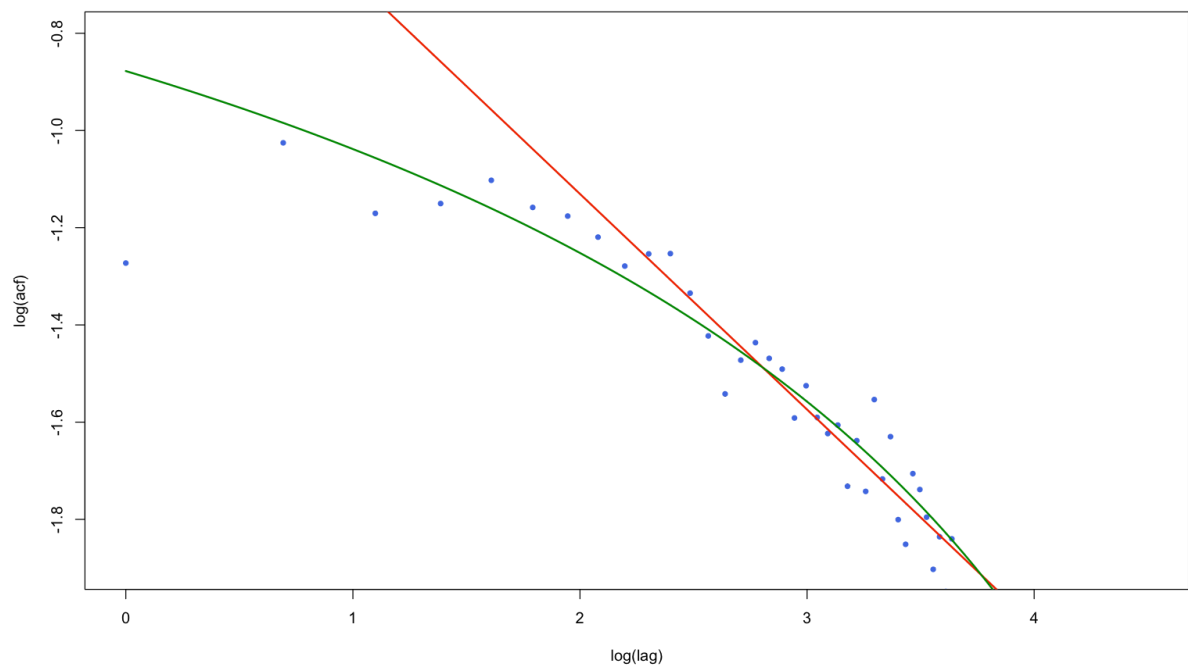


Figure 17: The red line is the conventional linear fit; the green line is the rough volatility prediction (2) .

Estimated mean-reversion time

According to the Garnier-Sølna formula,

$$\rho(\Delta) \sim 1 - \frac{1}{\Gamma(1 + 2H)} \left(\frac{\Delta}{\epsilon} \right)^{2H}.$$

so

$$\log(1 - \rho(\Delta)) \sim -\log \Gamma(1 + 2H) + 2H \log \Delta - 2H \log \epsilon = a + b \log \Delta.$$

Thus $2H = b$ and

$$\log \epsilon = -\frac{a}{b} + \frac{\log \Gamma(1 + b)}{b}.$$

This gives

```
In [35]: (log.eps <- -a/b + log(gamma(1+b))/b)
```

(Intercept): 4.85933753624717

```
In [36]: (eps <- exp(log.eps))
```

(Intercept): 128.93875655566

In other words, a mean reversion scale of 132 business days.

Log returns of VIX

VIX can be thought of as a measure of volatility smoothed over one month. Let's look at the distribution of VIX log-returns.

```
In [37]: vix <- spxVixData[, "Cl.VIX"]
spx <- spxVixData[, "Cl.GSPC"]

retVIX <- as.numeric(diff(log(vix))[-1])
retSPX <- as.numeric(diff(log(spx))[-1])

sdVIX <- as.numeric(sd(retVIX))
```

```
In [38]: hist(retVIX, breaks = 100, freq=F)
scale <- function(nu){sqrt(nu/(nu-2))}
curve(dt(x*scale(4)/sdVIX, df=4)*scale(4)/sdVIX, from=-.3, to=.3, col="red2", lwd=2, add=T)
curve(dnorm(x, mean=0, sd=sdVIX), from=-.3, to=.3, col="royalblue", lwd=2, add=T)
```

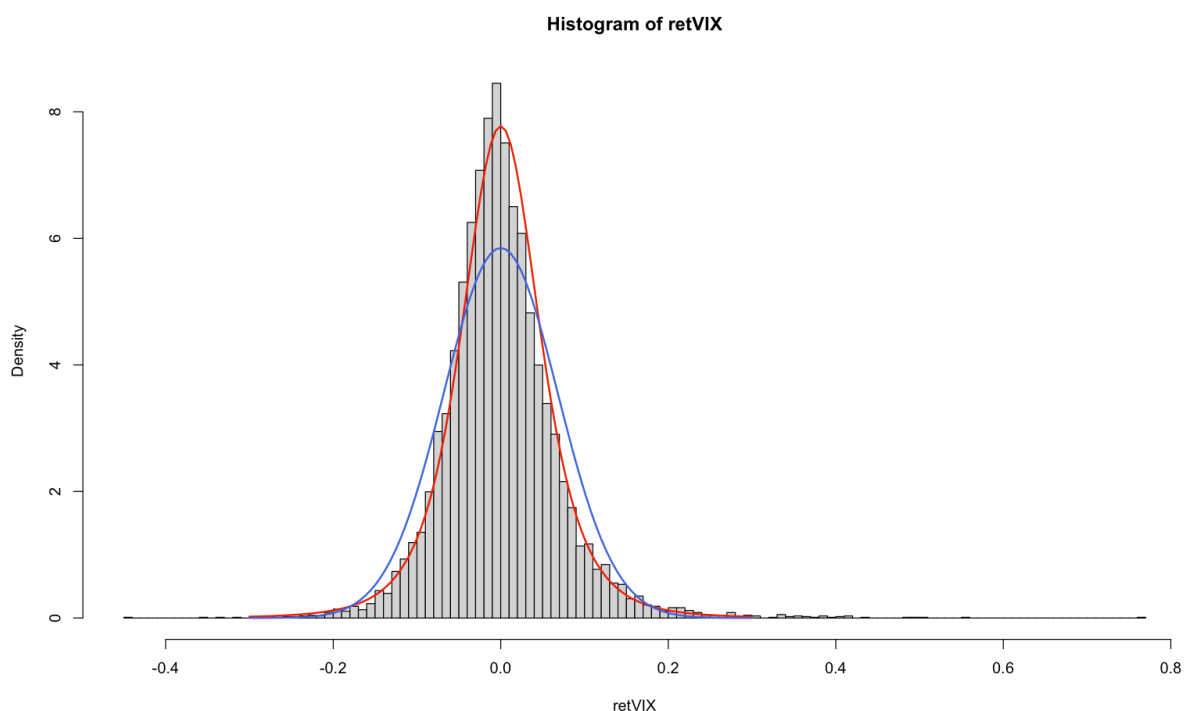


Figure 18: Log-returns of VIX are somewhat less fat-tailed than log-returns of SPX.

VIX vs SPX

We now regress log-returns of VIX against log-returns of SPX.

```
In [39]: fit.spxvix <- lm(retVIX~retSPX)

print(cor(retVIX,retSPX)) # Gets correlation
```

```
[1] -0.7142492
```

```
In [40]: # Scatter plot + fit
plot(retSPX,retVIX,xlab="SPX log returns",ylab="VIX log returns",pch=20,col=
abline(fit.spxvix,col="red2",lwd=2)
```

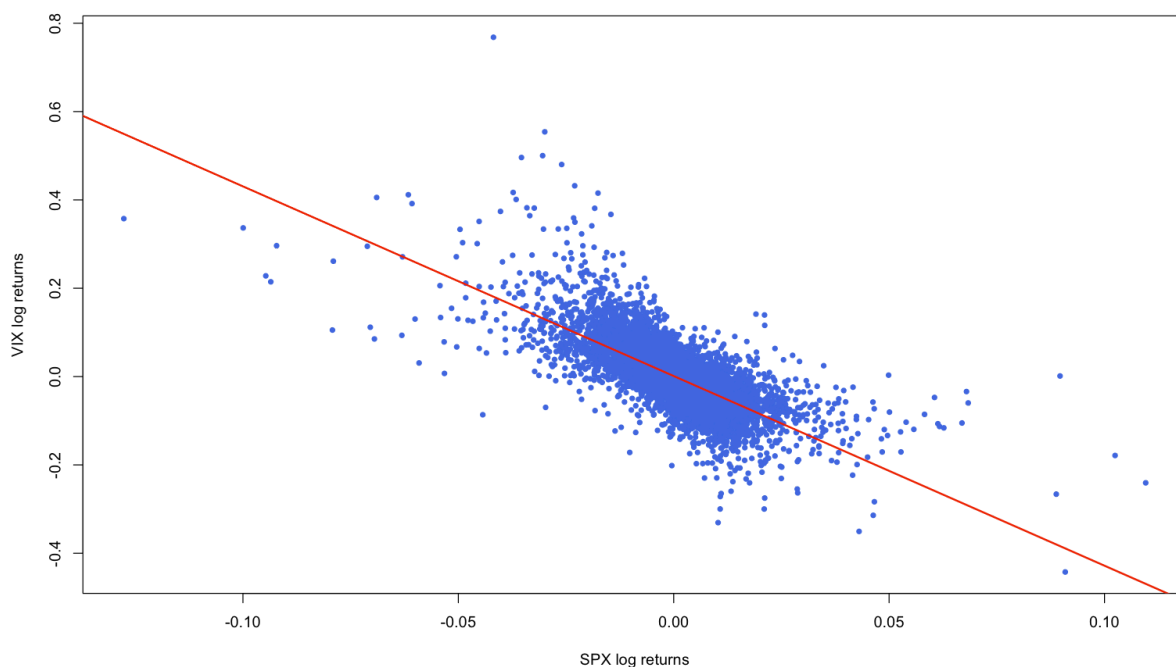


Figure 19: Regression of VIX log-returns vs SPX log-returns.

- Note the negative correlation ≈ -0.7 - similar to the correlation implied from fits of stochastic volatility models to option prices.

VIX vs SPX

Now just plot VIX vs SPX.

```
In [41]: n <- length(spx)
mycol <- hcl.colors(n = n, palette = "Batlow")
x <- as.numeric(spx)
y <- as.numeric(vix)
```

```
In [42]: plot(x[1],y[1],xlab="SPX",ylab="VIX",pch=20,cex=.5, col=mycol[1],
xlim=range(x,na.rm=T),ylim=range(y,na.rm=T))
for (i in 2:n){
  points(x[i],y[i],col=mycol[i],pch=20)
}
```

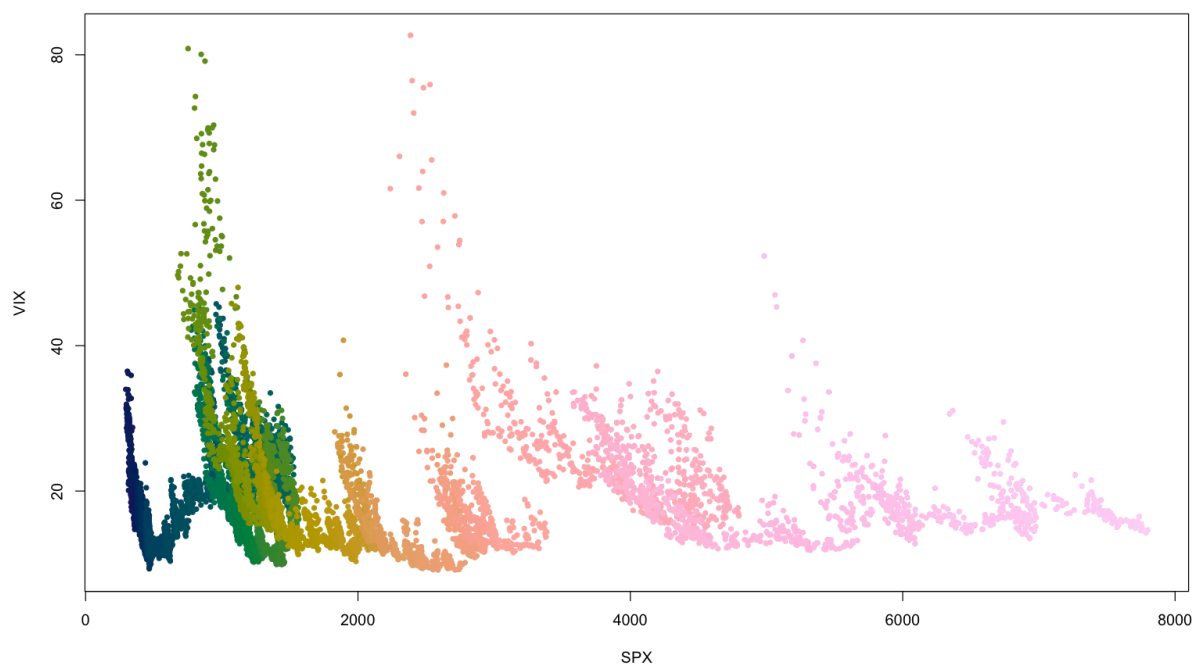


Figure 20: We see evidence of the path dependence of VIX. Points are colored according to date.

Volatility surface stylized properties

Having studied some properties of financial time series, let's now look at some stylized properties of volatility surfaces.

Smiles as of 15-Sep-2005

We see that the graph of implied volatility vs log-strike looks like a skewed smile. Some people say "smirk".



No description has been provided for this image

Figure 21: SPX volatility smiles as of September 15, 2005.

Smiles as of 02-Jul-2025

Now there are 54 expirations, not just 8!

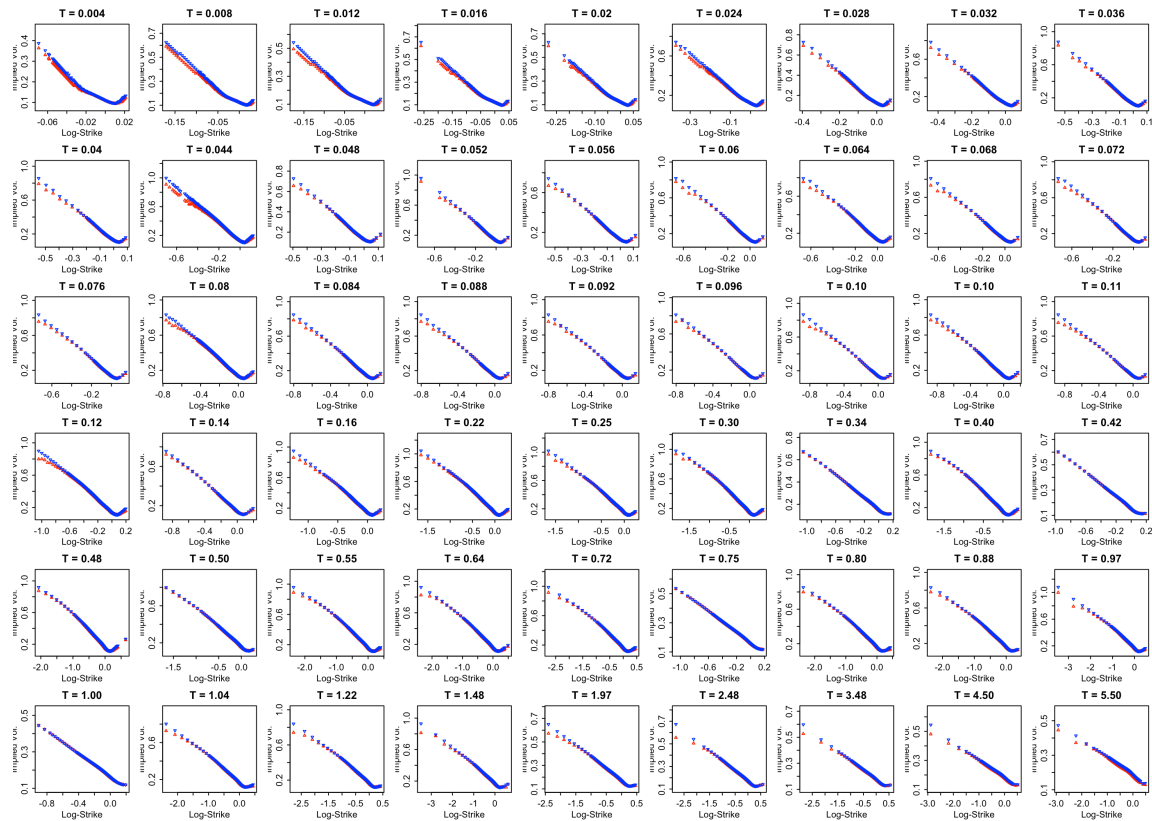


Figure 22: SPX volatility smiles as of July 02, 2025.

ODTE options

- ODTE contracts make up roughly 60% to 66% of all S&P 500 Index options volume.
- Across the entire U.S. equity market, they account for nearly a quarter of all listed volume. Expanding Footprint:
- Trading has shifted from just indexes into high-liquidity mega-cap stocks like Tesla, Nvidia, and Apple.

Why People Trade Them

- Premiums are incredibly cheap, yielding massive directional leverage for low capital.
- Positions are opened and closed within the same day, completely avoiding overnight global news shocks.
- Institutions use them to hedge highly specific intraday events, like Fed rate calls or CPI releases, at low cost.
- Option sellers exploit accelerated collapse of option premium close to 4pm.

ODTE option prices as of July 02, 2025

The following dataset consists of all options as of Wednesday 02-Jul-2025 at 3pm expiring at 4pm on the same day.

```
In [43]: load("dat_20250702_1500.rData")
```

How many options have a bid of 5 cents?

```
In [44]: opt_data <- dat_20250702_1500
```

```
In [45]: opt_data[(opt_data$bid==.05),]
```

A data.frame: 12 × 5

	quote_datetime	strike	option_type	bid	ask
	<chr>	<dbl>	<chr>	<dbl>	<dbl>
17297445	2025-07-02 15:00:00	6120	P	0.05	0.1
17298255	2025-07-02 15:00:00	6125	P	0.05	0.1
17299065	2025-07-02 15:00:00	6130	P	0.05	0.1
17299875	2025-07-02 15:00:00	6135	P	0.05	0.1
17300685	2025-07-02 15:00:00	6140	P	0.05	0.1
17301495	2025-07-02 15:00:00	6145	P	0.05	0.1
17302305	2025-07-02 15:00:00	6150	P	0.05	0.1
17317290	2025-07-02 15:00:00	6245	C	0.05	0.1
17318100	2025-07-02 15:00:00	6250	C	0.05	0.1
17318910	2025-07-02 15:00:00	6255	C	0.05	0.1
17319720	2025-07-02 15:00:00	6260	C	0.05	0.1
17320530	2025-07-02 15:00:00	6265	C	0.05	0.1

Estimate ATM volatility

```
In [46]: opt_data[(opt_data$strike>6215)&(opt_data$strike<6230),]
```

A data.frame: 4 × 5

	quote_datetime	strike	option_type	bid	ask
	<chr>	<dbl>	<chr>	<dbl>	<dbl>
17313240	2025-07-02 15:00:00	6220	C	4.50	4.6
17313645	2025-07-02 15:00:00	6220	P	2.70	2.8
17314050	2025-07-02 15:00:00	6225	C	1.95	2.0
17314455	2025-07-02 15:00:00	6225	P	5.10	5.3

We estimate $\sigma\sqrt{T}$ by solving

$$\frac{\sigma\sqrt{T}}{\sqrt{2\pi}} \approx \frac{2.35}{6222}.$$

This gives

```
In [47]: (sig <- 2.35/6222*sqrt(2*pi))
```

```
0.000946733597779307
```

This represents roughly a one standard deviation move until expiration in one hour.

Far out-of-the-money options are 5 cents bid

- We see that there are 12 puts and calls expiring in one hour with a bid of 5 cents.
 - These options are PM-settled and so expire at 16:00.
- SPX was trading at roughly 6,223.
- A put struck at 6,120, 103 points out-of-the-money was 5 cents bid.
- A call struck at 6265, 42 points out-of-the-money was 5 cents bid.

```
In [48]: c(103/6222/sig, 42/6222/sig)
```

```
17.4855552686585 · 7.13003224547241
```

```
In [49]: c(pnorm(-17.5), pnorm(-7.1))
```

```
7.16345876623504e-69 · 6.23784446333159e-13
```

Inconsistency with diffusion

- The probability of a normally distributed variable making such moves is zero (to very many decimal places).
 - And these 5 cent bids are only bids; one might suppose that actual trades would take place somewhere between the bid and the offer.
- Conventional diffusion models just can't generate the size of moves over very short timescales that would be required to justify such option prices.
 - We would need to add jumps.
 - Rough volatility models can! In this sense, rough volatility models are jump-like.

Short-covering as an explanation

- High bids for options that would require an extreme move to end up in-the-money reflect the extreme short-end smile in the SPX market just prior to expiration.
- From the perspective of a trader, the explanation is straightforward: large moves do sometimes occur and it makes economic sense to bid for out-of-the-money options – at the very least to cover existing risk.
- A similar explanation points out that a position of short very out-of-the-money options requires margin. Covering the short fixes this.

Interpreting the smile

Given a volatility smile for a particular time to expiration, there is very little that can be deduced about the underlying dynamics:

- The realized variance (quadratic variation) of the underlying from now to expiration is uncertain.
- As the underlying price decreases, volatility tends to increase (anticorrelation of underlying moves and volatility moves).

3D plot

Interpolating by time to expiration, we obtain the following picture of the SPX volatility surface as of the close on September 15, 2005. $k := \log K/F$ is the log-strike and t is time to expiry in years.

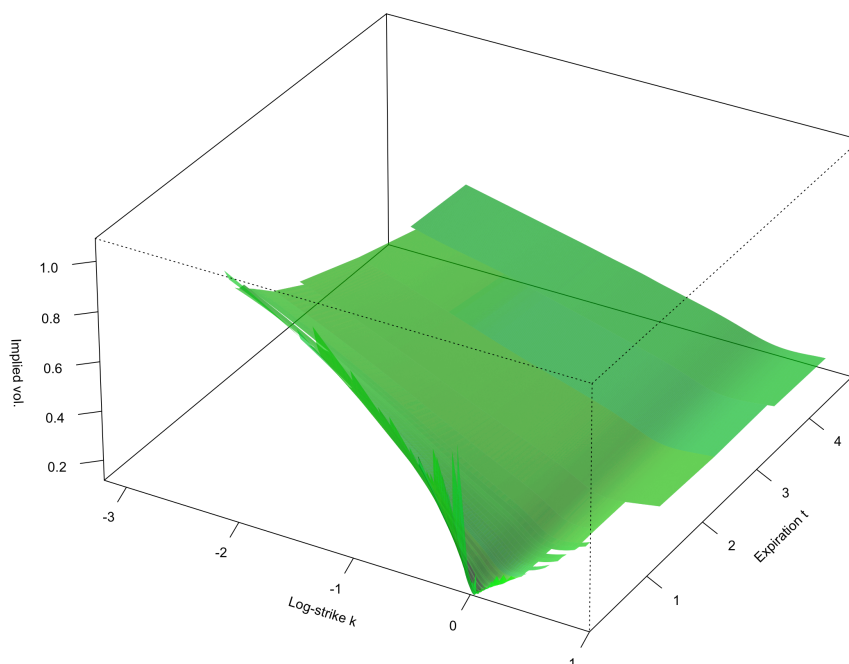


Figure 23: The SPX volatility surface as of September 15, 2005.

At-the-money (ATM) volatility skew

For a given time to expiration τ , we define the ATM volatility skew

$$\mathcal{S}(\tau) = \left. \frac{\partial}{\partial k} \sigma_{BS}(k, \tau) \right|_{k=0}.$$

It turns out that the observed term structure of ψ is consistent with only a very restricted class of stochastic volatility models, and is in fact inconsistent with all conventional stochastic volatility models, including those with jumps.

Observed term structure of ATM volatility skew

We study a period of history over which the ATM skew was relatively stable.

```
In [50]: # Download .zip from GitHub

release_url <- "https://github.com/jgatheral/MTH9875/raw/main/9875_1_2026.zip"
local_zip <- "9875-1.zip"

download.file(release_url, local_zip, mode = "wb")
unzip(local_zip)
```

```
In [51]: load("volSkewCurv20230831.rData")
```

```
In [52]: tail(names(vsc.list))
```

```
'20230824' · '20230825' · '20230828' · '20230829' · '20230830' · '20230831'
```

```
In [53]: names(vsc.list[['20230831']])
```

```
'expiries' · 'atmVol' · 'atmSkew' · 'atmCurv'
```

```
In [54]: m0 <- 3816
vsl <- vsc.list[m0:(m0+85)]
n <- length(names(vsl))
mycol <- hcl.colors(n = n, palette = "Batlow")
cex <- .5

c(names(vsl)[1], names(vsl)[64])
```

```
'20110301' · '20110531'
```

```
In [55]: plot(vsl[[1]]$expiries, abs(vsl[[1]]$atmSkew), col=mycol[1], pch=20, cex=cex, ylab=expression(psi(tau)),
             xlab=expression(paste("Expiration ", tau)), ylab=expression(psi(tau)))
for (i in 2:n){
  tryCatch(points(vsl[[i]]$expiries, abs(vsl[[i]]$atmSkew), col=mycol[i], pch=20, cex=cex),
            error=function(e){})
}
```

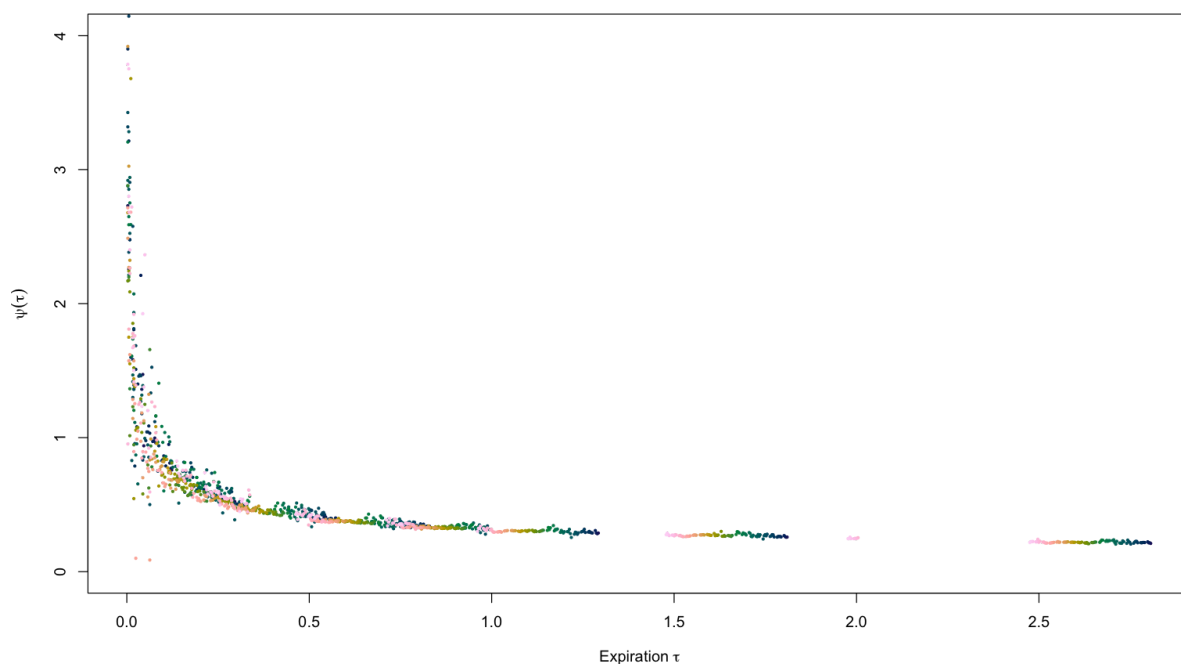


Figure 24: Decay of ATM skew with respect to time to expiration. Data is SPX from 03-Mar-2011 to 31-May-2011. Points are colored according to date.

Log-log plot of empirical ATM skew

```
In [56]: lvsl.expiries <- log(vsl[[1]]$expiries)
lvsl.atmSkew <- log(abs(vsl[[1]]$atmSkew))

for (i in 2:n){
  lvsl.expiries <- c(lvsl.expiries, log(vsl[[i]]$expiries))
  lvsl.atmSkew <- c(lvsl.atmSkew, log(abs(vsl[[i]]$atmSkew)))
}
```

```
In [57]: pick <- (lvsl.expiries > -3)
summary(fit.lm <- lm(lvsl.atmSkew[pick] ~ lvsl.expiries[pick]))
```

Call:

```
lm(formula = lvsl.atmSkew[pick] ~ lvsl.expiries[pick])
```

Residuals:

Min	1Q	Median	3Q	Max
-2.38499	-0.04047	-0.00531	0.03406	0.56387

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1.139919	0.004207	-270.9	<2e-16 ***
lvsl.expiries[pick]	-0.390953	0.003553	-110.0	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1137 on 1079 degrees of freedom

Multiple R-squared: 0.9182, Adjusted R-squared: 0.9181

F-statistic: 1.211e+04 on 1 and 1079 DF, p-value: < 2.2e-16

```
In [58]: plot(lvsl.expiries, lvsl.atmSkew, col="red2", pch=20, cex=0.5,
  xlab=expression(paste("log ", tau)), ylab=expression(paste("log ", psi(tau))),
  abline(fit.lm, col="royalblue", lwd=2))
```

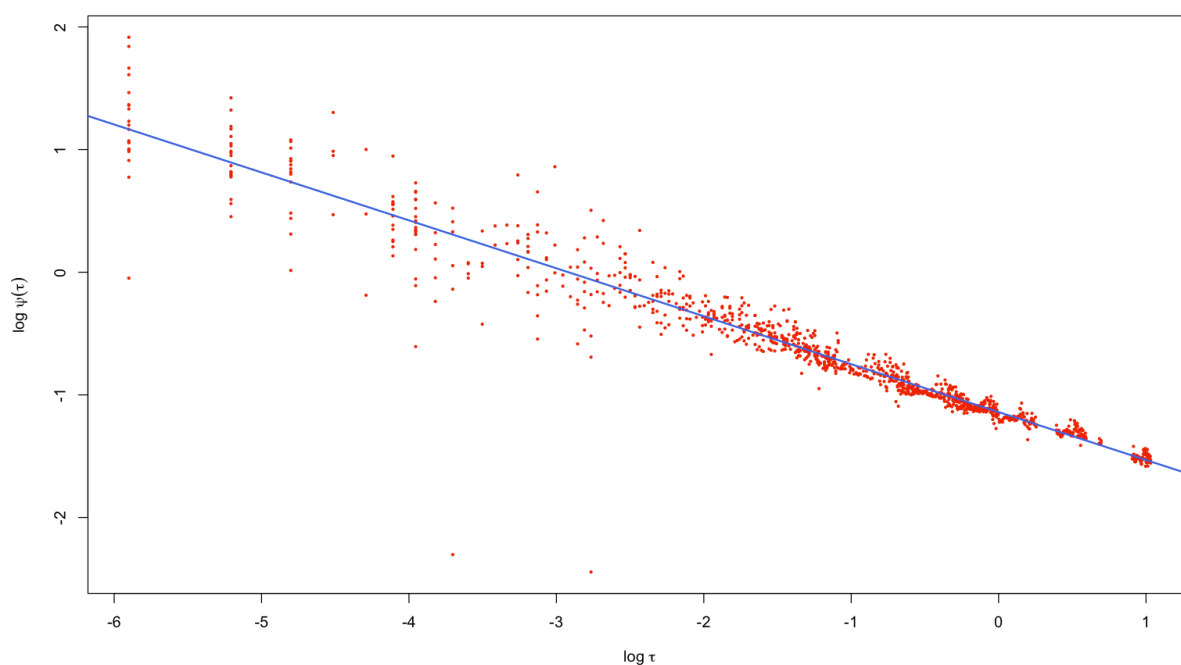


Figure 25: Log-log plot of decay of ATM skew (red dots) with respect to time to expiration. Power-law fit $\psi(\tau) \sim \tau^{-.39}$ in blue. Data is SPX from 03-Mar-2011 to 31-May-2011.

Plot of ATM skew with power-law fit

```
In [59]: plot(vsl[[1]]$expiries,abs(vsl[[1]]$atmSkew),col="red2",pch=20,cex=cex,xlim=
          xlab=expression(paste("Expiration ",tau)),ylab=expression(psi(tau)))
for (i in 2:n){
  points(vsl[[i]]$expiries,abs(vsl[[i]]$atmSkew),col="red2",pch=20,cex=cex
}

a <- fit.lm$coef[1]; b <- fit.lm$coef[2]
curve(exp(a+b*log(x)),from=0,to=3,col="royalblue",add=T,n=5000,lwd=2)
```

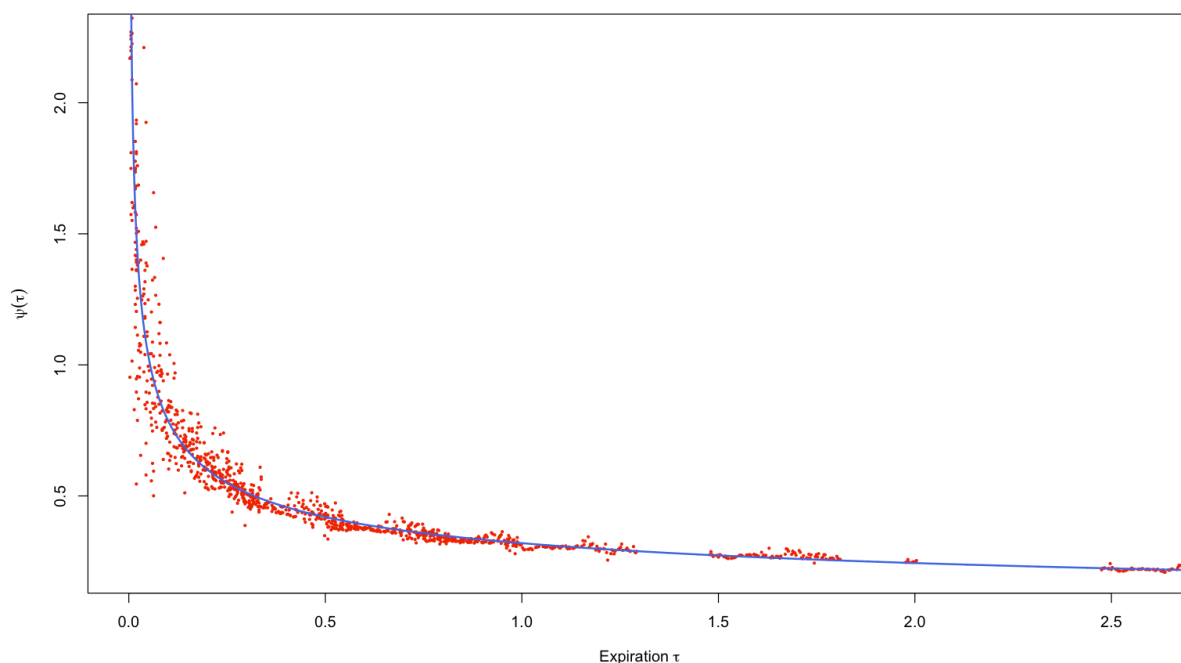


Figure 26: Decay of ATM skew (red dots) with respect to time to expiration. Data is SPX from 03-Mar-2011 to 31-May-2011.

More recently

```
In [60]: m0 <- 6897
# m0 <- 6780
vsl <- vsc.list[m0:(m0+41)]
n <- length(names(vsl))
mycol <- hcl.colors(n = n, palette = "Batlow")

c(names(vsl)[1],names(vsl)[41])
```

'20230601' · '20230731'

Log-log plot of empirical ATM skew

```
In [61]: plot(log(vsl[[1]]$expiries), log(abs(vsl[[1]]$atmSkew)), col=mycol[1], pch=20, cex=0.3,
           xlab=expression(paste("Expiration ", tau)), ylab=expression(psi(tau)))
for (i in 2:n){
  tryCatch(points(log(vsl[[i]]$expiries), log(abs(vsl[[i]]$atmSkew)),
                 col=mycol[i], pch=20, cex=.3), error=function(e){e})
}
```

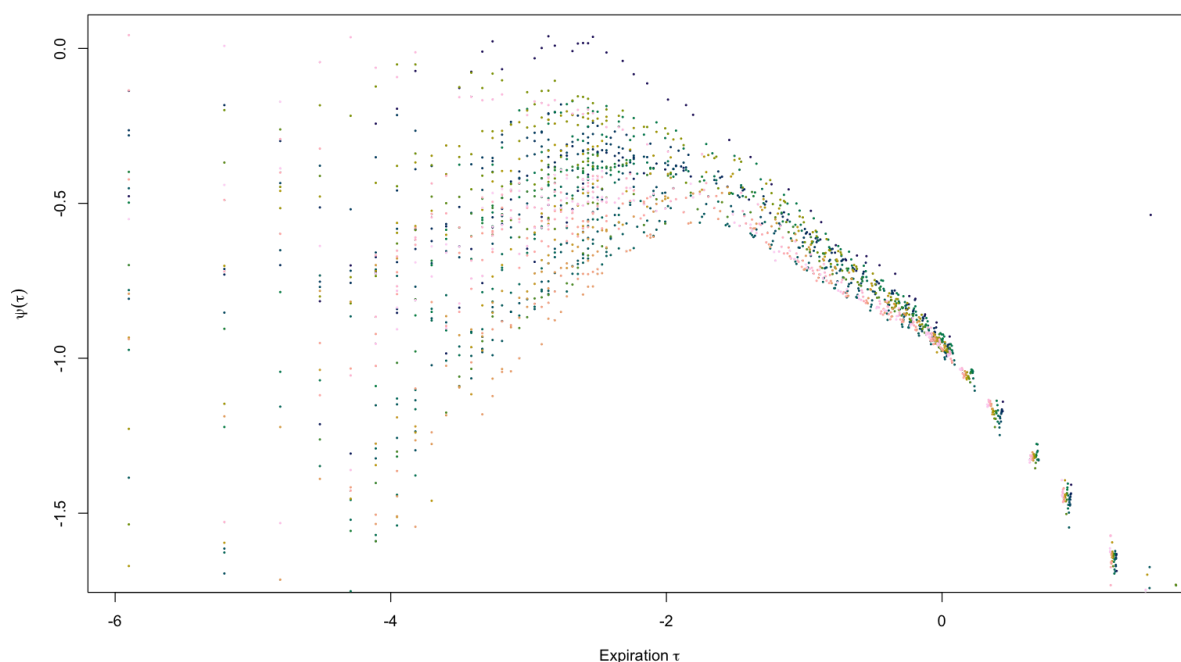


Figure 27: Log-log plot of decay of ATM skew with respect to time to expiration. Data is SPX from 01-Jun-2023 to 31-Jul-2023. Points are colored according to date. The pattern of the points looks like two power laws!

Two power-law fits

```
In [62]: lvsl.expiries <- log(vsl[[1]]$expiries)
lvsl.atmSkew <- log(abs(vsl[[1]]$atmSkew))

for (i in 2:n){
  lvsl.expiries <- c(lvsl.expiries, log(vsl[[i]]$expiries))
  lvsl.atmSkew <- c(lvsl.atmSkew, log(abs(vsl[[i]]$atmSkew)))
}
```

```
In [63]: pick <- (lvsl.expiries > 0)
summary(fit.lm1 <- lm(lvsl.atmSkew[pick] ~ lvsl.expiries[pick]))
```

```
Call:
lm(formula = lvsl.atmSkew[pick] ~ lvsl.expiries[pick])

Residuals:
    Min       1Q   Median       3Q      Max
-0.32226 -0.01778 -0.00207  0.01492  1.25728

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   -0.951301   0.008246  -115.4  <2e-16 ***
lvsl.expiries[pick] -0.557025   0.008440   -66.0  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08593 on 313 degrees of freedom
Multiple R-squared:  0.933,    Adjusted R-squared:  0.9327
F-statistic: 4356 on 1 and 313 DF,  p-value: < 2.2e-16
```

```
In [64]: pick <- (lvsl.expiries < -.5)&(lvsl.expiries > -2)
```

```
In [65]: summary(fit.lm2 <- lm(lvsl.atmSkew[pick] ~ lvsl.expiries[pick]))
```

```
Call:
lm(formula = lvsl.atmSkew[pick] ~ lvsl.expiries[pick])

Residuals:
    Min       1Q   Median       3Q      Max
-0.167985 -0.049781 -0.003984  0.040982  0.209288

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   -0.919229   0.008231  -111.68  <2e-16 ***
lvsl.expiries[pick] -0.283737   0.006749   -42.04  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.0621 on 459 degrees of freedom
Multiple R-squared:  0.7939,    Adjusted R-squared:  0.7934
F-statistic: 1768 on 1 and 459 DF,  p-value: < 2.2e-16
```

```
In [66]: plot(lvsl.expiries,lvsl.atmSkew,col="red2",pch=20,cex=0.5,
              xlab=expression(paste("log ",tau)),ylab=expression(paste("log ",psi(ta
              abline(fit.lm1,col="royalblue",lwd=2)
              abline(fit.lm2,col="royalblue",lwd=2)
```

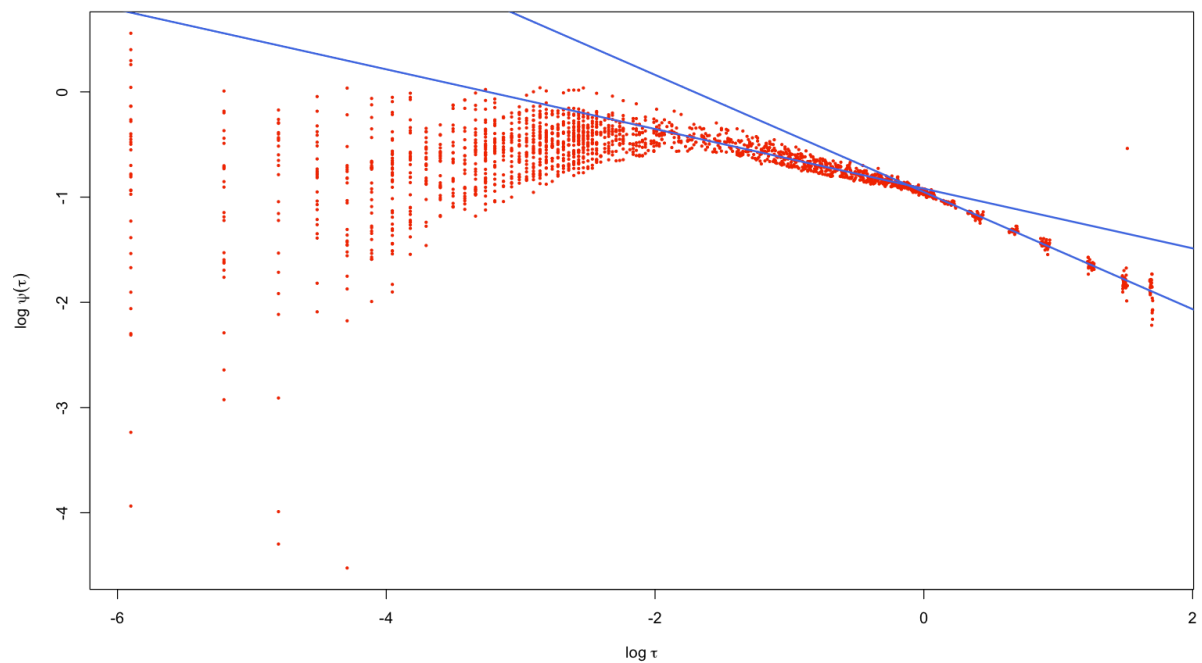


Figure 28: Log-log plot of decay of ATM skew with respect to time to expiration. Data is SPX from 01-Jun-2023 to 31-Jul-2023. Superimposed in blue are the two power-law fits: $\tau^{-.28}$ and $\tau^{-.56}$, respectively.

Remark on the two power-laws

- By changing dates, you can check that this two power-law behavior seems to be persistent.
 - We used to have only one power law but now it seems that we have two!
- This is consistent with observations by De Marco et al. ^[6]

Skew term structure is not always power-law

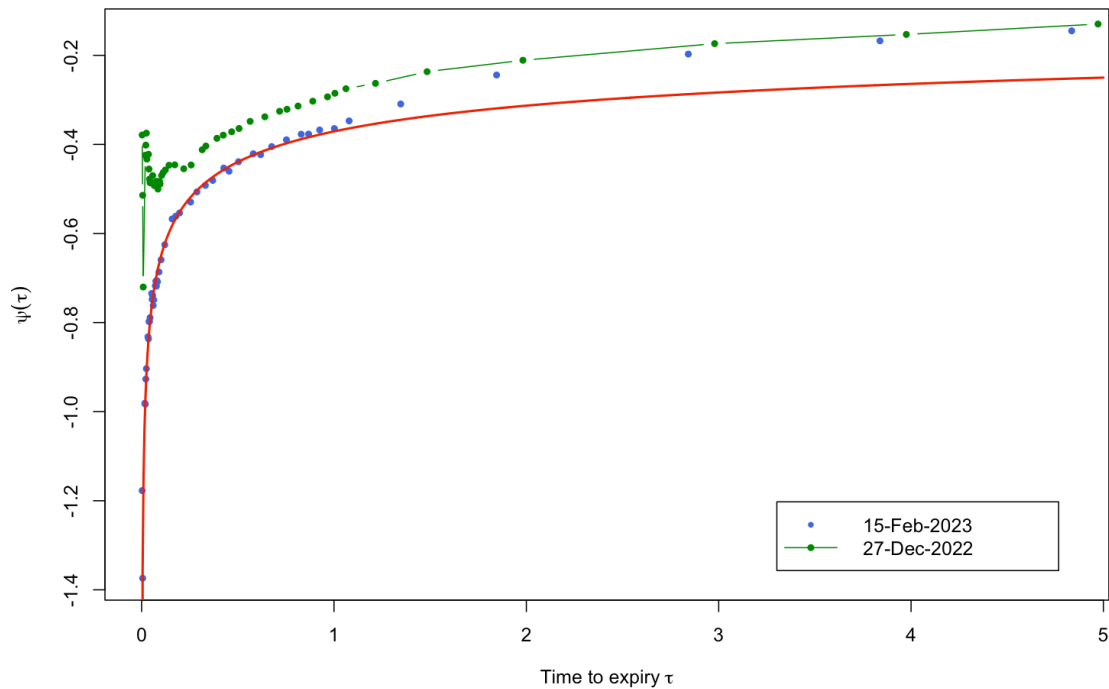


Figure 29. ATM skew term structure on two different dates. On 27-Dec-2022, the skew term structure is not even monotonic!

Total variance plot

- Define the implied total variance $w(k, \tau) := \sigma_{BS}(k, \tau)^2 \tau$.
- To avoid calendar spread arbitrage, we must have $w(k, \tau)$ non-decreasing in τ for fixed k .
 - If lines on a total variance plot cross, there is calendar spread arbitrage.
- The non-monotonic skew term structure on 27-Dec-2022 leads one to suspect calendar spread arbitrage.

Total variance plot as of 27-Dec-2022

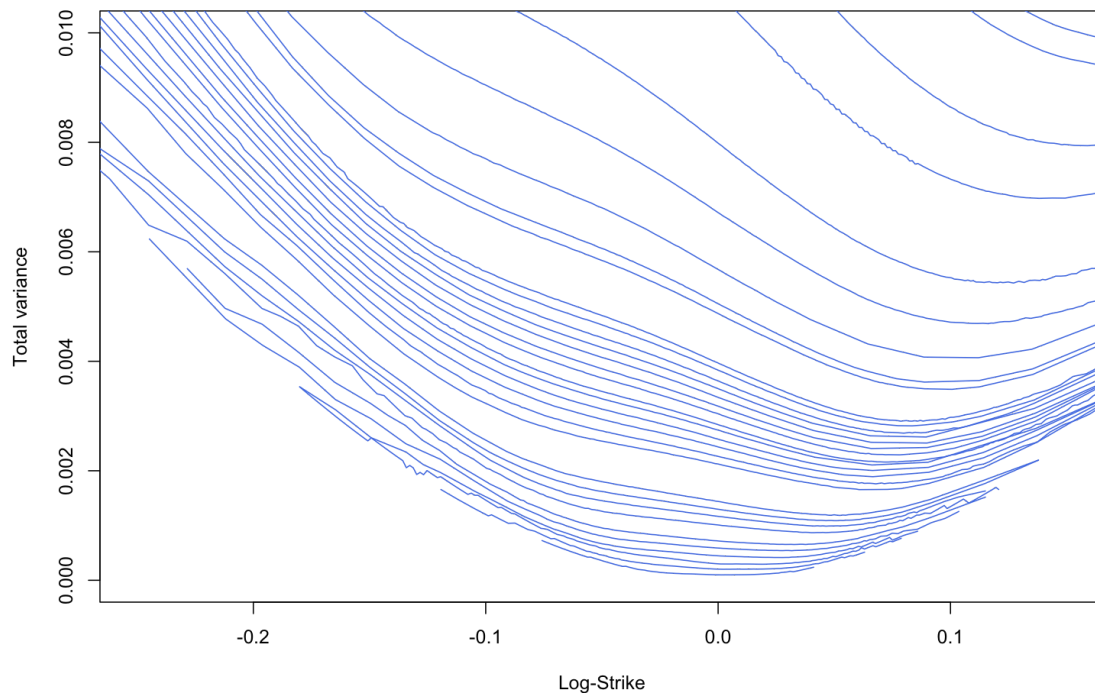


Figure 30. On 27-Dec-2022, no calendar spread arbitrage. However, some individual total variance curves are W-shaped.

Scaling of total variance

- The rough SABR formula [\[7\]](#) suggests that we should have

$$\frac{w(k, \tau)}{w(0, \tau)} \approx f \left(\tau^{-\gamma} \frac{k}{\Sigma_{BS}(0)} \right).$$

- Roughly speaking, total variance curves should scale as a power-law.
- Figure 31. does suggest close-to-power-law scaling, even in the 27-Dec-2022 case.

Scaling of total variance

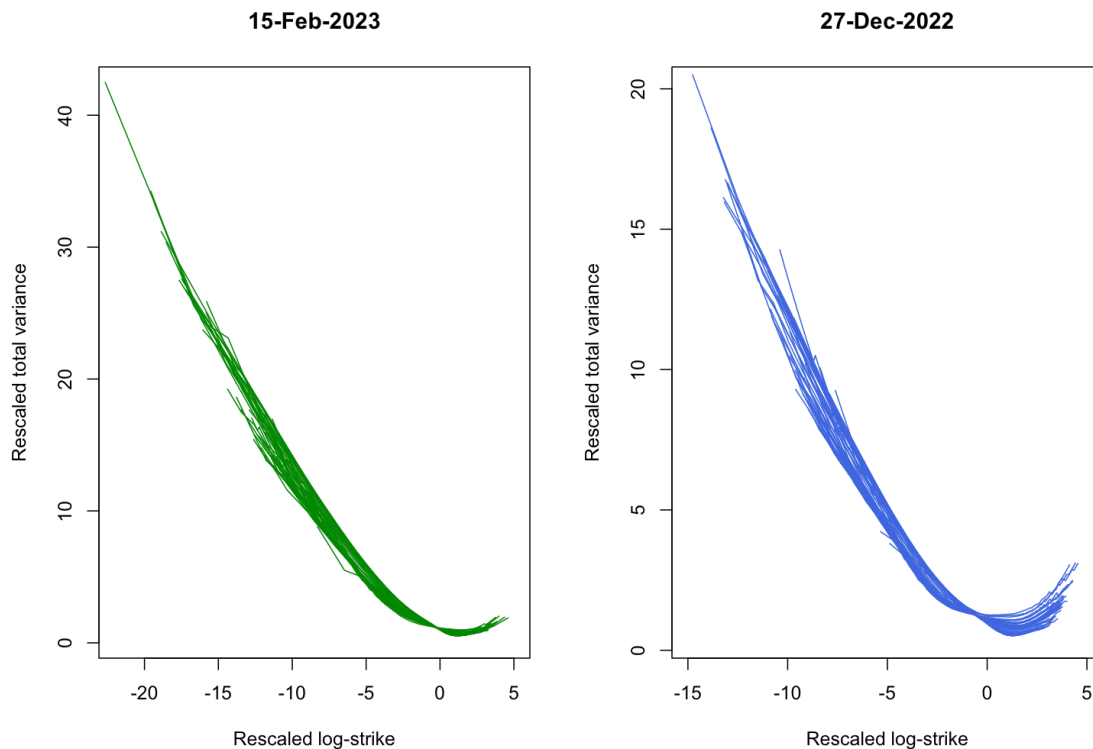


Figure 31. ATM skew term structure on two different dates. On 27-Dec-2022, the skew term structure is not even monotonic!

Scaling of total variance for ODTE options

- We can use our ODTE data from July 2, 2025 to investigate the scaling of smiles as options are about to expire.
- `dte0_smiles_20250702.rData` has a list of all smiles for that day.

```
In [67]: load("dte0_smiles_20250702.rData")
```

```
In [68]: source("black_scholes.R")
         source("plot_ivols.R")
```

```
In [69]: head(dte0_smiles[[330]])
```

A data.frame: 6 × 7

	QDT	Texp	Strike	Bid	Ask	Fwd	CallMid
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
132	2025-07-02 15:00:00	0.004	6120	0.09887901	0.10684011	6221.796	101.87139
133	2025-07-02 15:00:00	0.004	6125	0.09448640	0.10216135	6221.796	96.87139
134	2025-07-02 15:00:00	0.004	6130	0.09007777	0.09746347	6221.796	91.87139
135	2025-07-02 15:00:00	0.004	6135	0.08565182	0.09274495	6221.796	86.87139
136	2025-07-02 15:00:00	0.004	6140	0.08120709	0.08800406	6221.796	81.87139
137	2025-07-02 15:00:00	0.004	6145	0.07674193	0.08323882	6221.796	76.87139

```
In [70]: plot_dte0 <- function(ivolData, plot = TRUE){
  include <- !is.na(as.numeric(ivolData$Bid))
  ivolData <- ivolData[include, ]

  bidVols <- as.numeric(ivolData$Bid)
  askVols <- as.numeric(ivolData$Ask)
  midVols <- (bidVols + askVols) / 2

  t <- unique(ivolData$Texp)
  fwd <- unique(ivolData$Fwd)
  k <- log(ivolData$Strike / ivolData$Fwd)
  qdt <- unique(ivolData$QDT)

  volInterp <- function(xout) stinter(x = k, y = midVols, xout)$y

  sig <- tryCatch(volInterp(0), error = function(e) NA_real_) * sqrt(t)
  atmVol <- tryCatch(volInterp(0),
    error = function(e) NA_real_)
  atmSkew <- tryCatch((volInterp(sig / 10) - volInterp(-sig / 10)) / (2 * sig),
    error = function(e) NA_real_)
  atmCurv <- tryCatch((volInterp(sig / 10) + volInterp(-sig / 10) - 2 * atmVol),
    error = function(e) NA_real_)

  if (plot) {
    ybottom <- 0.8 * min(bidVols, na.rm = TRUE)
    ytop <- 1.2 * max(askVols, na.rm = TRUE)
    plot(k, bidVols,
      col = "red", pch = 24, cex = 0.5,
      xlim = c(min(k), max(k)), ylim = c(ybottom, ytop),
      main = qdt, xlab = "Log-Strike", ylab = "Implied Vol.")
    points(k, askVols, col = "blue", pch = 25, cex = 0.5)
  }
}
```

```
invisible(list(k = k, bidVols = bidVols, askVols = askVols, midVols = midVols,
             expiries = t, fwds = fwd,
             atmVol = atmVol, atmSkew = atmSkew, atmCurv = atmCurv))
}
```

The smile as of 15:00:00

```
In [71]: res.plot <- plot_dte0(dte0_smiles[[330]])
```

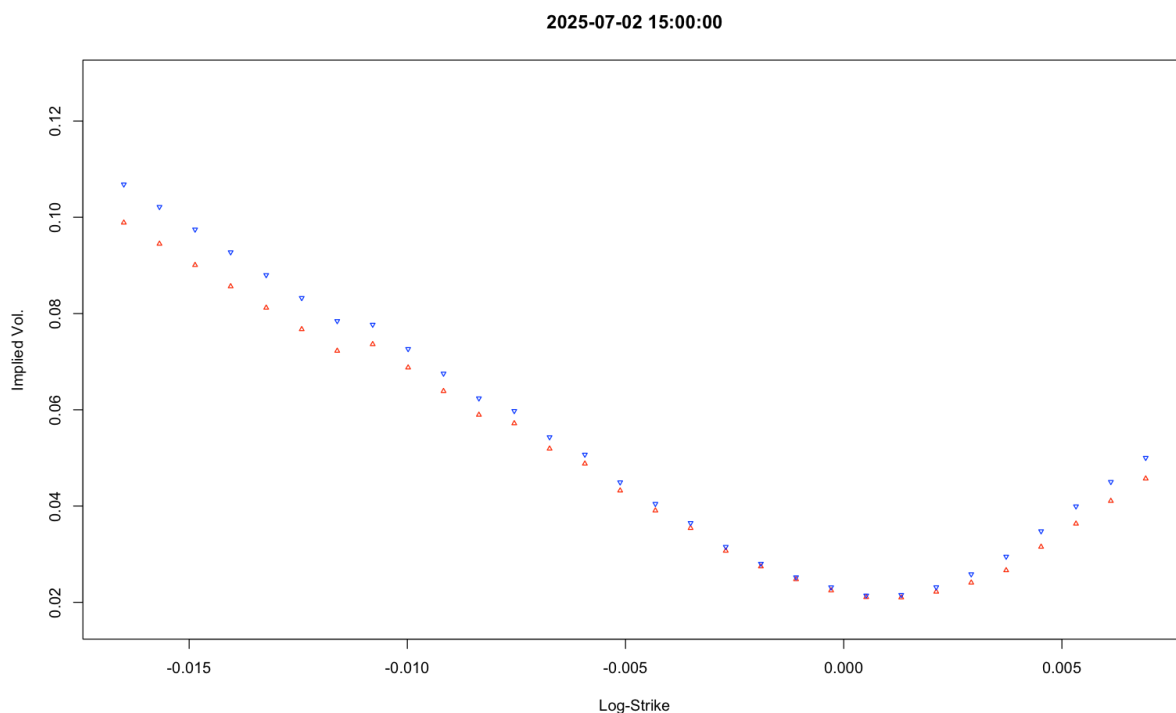


Figure 32: The ODTE smiles as of 3pm on July 2, 2025; expiry is 4pm the same day.

```
In [72]: plot_dte0_slices <- function(smiles_list, slices, col_palette = "Dark2") {
  results <- lapply(slices, function(i) plot_dte0(smiles_list[[i]], plot = F))

  if (is.character(col_palette) && length(col_palette) == 1) {
    cols <- hcl.colors(length(slices), palette = col_palette)
  } else {
    cols <- rep_len(col_palette, length(slices))
  }

  xrange <- range(unlist(lapply(results, `[`, "k")), na.rm = TRUE)
  yrange <- range(unlist(lapply(results, `[`, "midVols")), na.rm = TRUE)
  yrange <- c(0.8 * yrange[1], 1.2 * yrange[2])
  xrange <- c(-.01, .005)
  yrange <- c(0.02, .1)
  plot(NULL,
       xlim = xrange, ylim = yrange,
       xlab = "Log-Strike", ylab = "Implied Vol.",
       main = paste("DTE0 slices:", paste(slices, collapse = ", ")))

  for (j in seq_along(slices)) {
```

```

    lines(results[[j]]$k, results[[j]]$midVols, col = cols[j], lwd = 1.5)
  }

  legend("topright", legend = paste("Slice", slices),
        col = cols, lwd = 1.5, cex = 0.7)

  invisible(results)
}

```

Many smiles

In [73]: `plot_dte0_slices(dte0_smiles, c(50,100,150,200,250,300,350), col_palette = "`

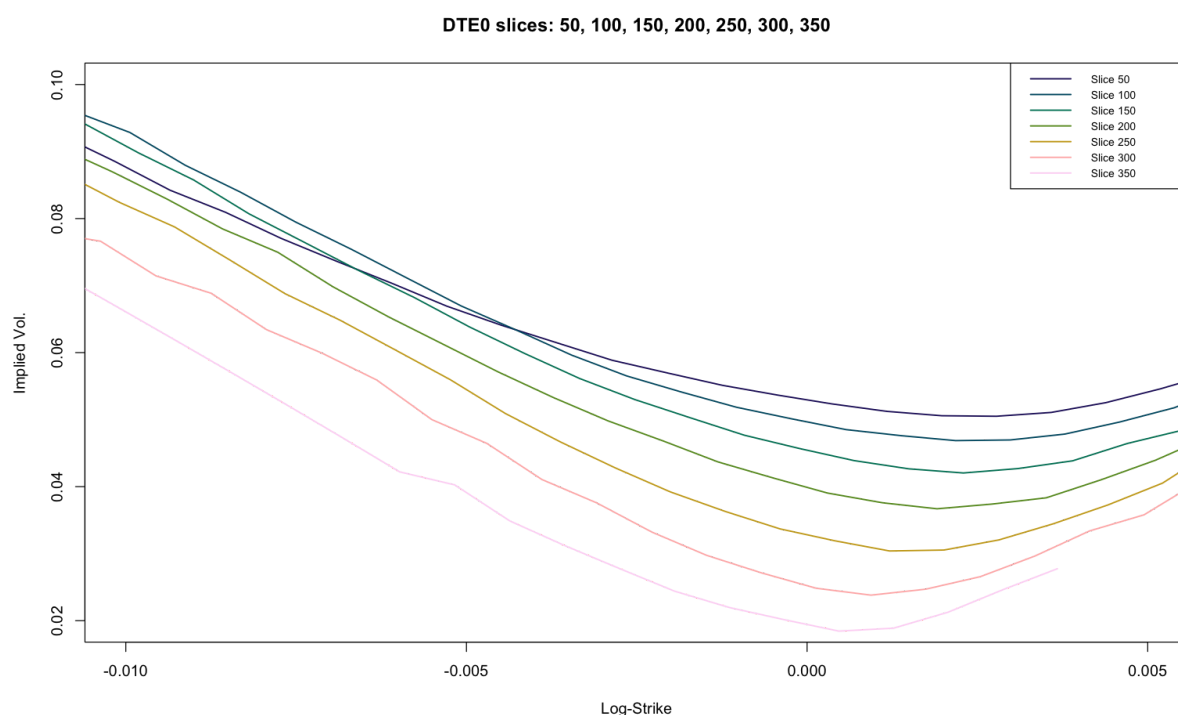


Figure 33: Keeping $T = 0.004$ implied vol. falls as time to expiration decreases.

Normalized vol. vs normalized sstrike

- We normalize the log-strike k :

$$k \mapsto z = \frac{k}{\sigma(k, T)\sqrt{T}}.$$

- We normalize the vol by dividing by ATM vol:

$$\sigma(k, T) \mapsto \frac{\sigma(k, T)}{\sigma(0, T)}.$$

In [74]: `plot_dte0_slices_normalized <- function(smiles_list, slices, col_palette = "`

```

dt <- smiles_list[[1]][[1]][[1]]

results <- lapply(slices, function(i) plot_dte0(smiles_list[[i]], plot = F

if (is.character(col_palette) && length(col_palette) == 1) {
  cols <- hcl.colors(length(slices), palette = col_palette)
} else {
  cols <- rep_len(col_palette, length(slices))
}

# Compute atmVol, normalized strike  $z = k / (atmVol * sqrt(t))$ , normalized
zlist <- vector("list", length(slices))
volnlist <- vector("list", length(slices))
for (j in seq_along(slices)) {
  r <- results[[j]]
  atmVol <- approx(x = r$k, y = r$midVols, xout = 0)$y
  zlist[[j]] <- r$k / (atmVol * sqrt(r$expiries)^.9)
  volnlist[[j]] <- r$midVols / atmVol
}

xrange <- range(unlist(zlist), na.rm = TRUE)
yrange <- range(unlist(volnlist), na.rm = TRUE)
yrange <- c(0.8 * yrange[1], 1.2 * yrange[2])

xrange <- c(-4,2)
yrange <- c(1,3)

plot(NULL,
      xlim = xrange, ylim = yrange,
      xlab = expression(k / (sigma[ATM] * sqrt(t))),
      ylab = expression(sigma / sigma[ATM]),
      main = paste(dt,"DTE0 slices:", paste(slices, collapse = ", ")))

for (j in seq_along(slices)) {
  lines(zlist[[j]], volnlist[[j]], col = cols[j], lwd = 2)
}

legend("topright", legend = paste("Slice", slices),
      col = cols, lwd = 2, cex = 0.7)

invisible(results)
}

```

In [75]: `plot_dte0_slices_normalized(dte0_smiles, c(50,100,150,200,250,300,350), col_`

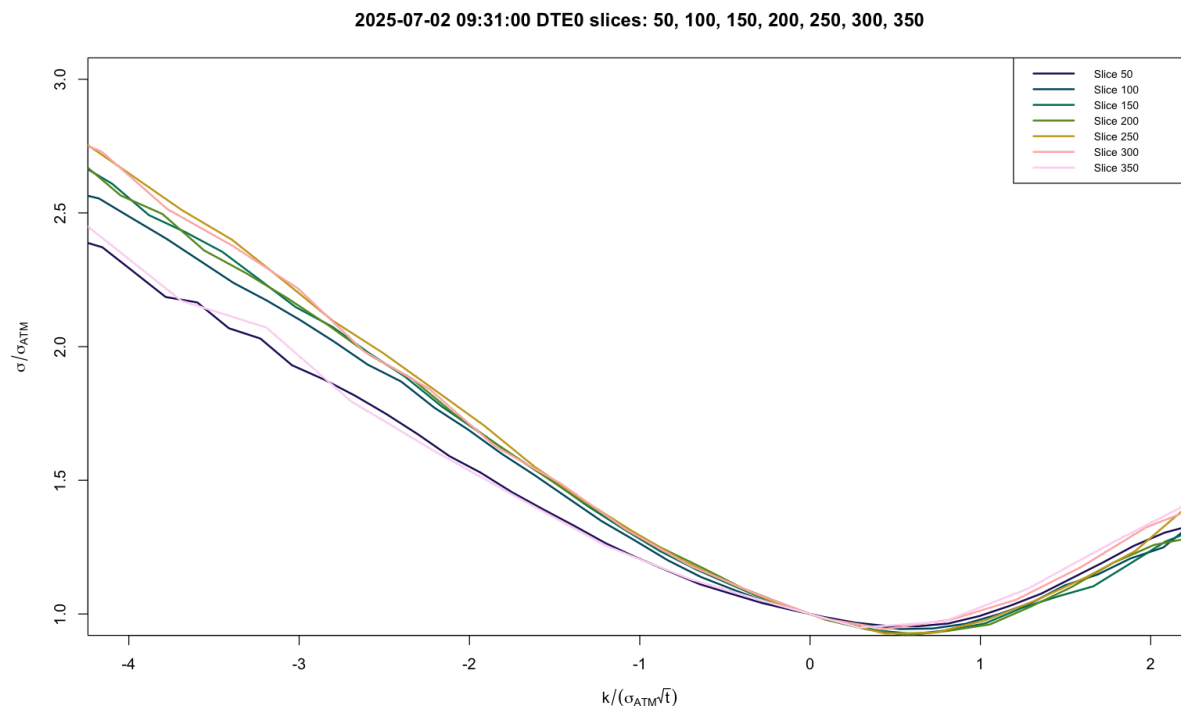


Figure 34: The smiles scale approximately as $1/\sqrt{T}$.

Volatility must be rough

- We see from Figure 34. that 0DTE smiles scale as a power-law T^γ with $\gamma \approx \frac{1}{2}$.
- According to Fukasawa^[6], if the asymptotic skew is power-law, volatility must be rough:

"... there is an arbitrage opportunity if volatility is not rough given an option market with volatility skew obeying the power law, under the assumption that the asset price is a positive continuous Itô semimartingale."

- Here we see that not only is the skew power-law, but the entire smile scales as a power-law as $T \rightarrow 0$.
 - Or there could be jumps, but that is another story...

Joint dynamics of ATM vol. and ATM skew

Here is a figure from a paper by Bourgey et al. ^[2]

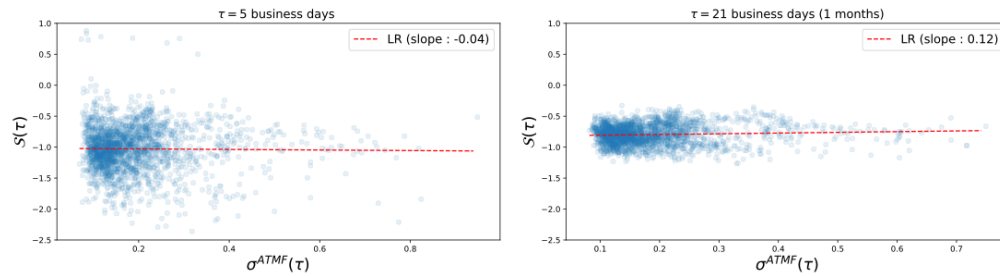


Figure 11: ATM skew w.r.t. ATM implied volatility for different time to maturities. The plots are generated using data reconstructed via the SVI model, so the time to maturity of n business days is often interpolated. The left plot corresponds to 5 days, while the right plot to 21 days.

Figure 35: The scatter plots show a very weak relationship between ATM skew and ATM vol. As we will see, this turns out to be roughly consistent with lognormal stochastic volatility models.

Option market making

- From the definition of implied volatility $\sigma_{BS}(k, T)$, the market price of an option is given by

$$C(S, K, T) = C_{BS}(S, K, T, \sigma_{BS}(k, T))$$

where C_{BS} denotes the Black-Scholes formula and $k = \log K/S$ is the log-strike.

- Market makers, when updating option prices using the Black-Scholes formula, typically consider two effects:
 - The explicit spot effect

$$\frac{\partial C}{\partial S} \delta S$$

- The change in implied volatility conditional on a change in the spot

$$\frac{\partial C}{\partial \sigma} \mathbb{E}[\delta \sigma | \delta S].$$

Estimating $\mathbb{E}[\delta \sigma(T) | \delta X]$

- ATM implied volatilities $\sigma_t(T) = \sigma_{BS,t}(0, T)$ and stock prices are both observable.
- Market makers can estimate the second component using a simple regression:

$$\delta \sigma_t(T) = \beta_t(T) \frac{\delta S_t}{S_t} + \text{noise} =: \beta_t(T) \delta X_t + \text{noise}.$$

- Then

$$\beta_t(T) = \frac{\mathbb{E}_t [d\langle \sigma(T), X \rangle_t]}{\mathbb{E}_t [d\langle X \rangle_t]}.$$

The skew-stickiness ratio

As before, for a given time to expiration T , we define the ATM volatility skew

$$\mathcal{S}_t(T) = \left. \frac{\partial}{\partial k} \sigma_{\text{BS}}(k, T) \right|_{k=0}.$$

- [Bergomi]^[2] calls

$$\mathcal{R}(T) = \frac{\beta(T)}{\psi(T)}$$

the *skew-stickiness ratio* or *SSR*.

Sticky strike and sticky delta

In the old days, traders would typically make one of two assumptions:

- **Sticky delta** where the ATM volatility is fixed.
 - In this case, when S increases to $S + \delta S$, $\delta\sigma(T) = 0$ so $\mathcal{R}(T) = 0$.

or

- **Sticky strike** where the implied volatility is fixed for a given strike independent of the stock price.
 - In this case, when S increases to $S + \delta S$,

$$\delta\sigma(T) = \sigma(S + \delta S) - \sigma(S) \approx \psi(T) \delta S$$

$$\text{so } \beta(T) = \psi(T) \text{ and } \mathcal{R}(T) = 1.$$

Listed options were thought of as sticky strike and OTC options as sticky delta.

SSR time series from Vola Dynamics

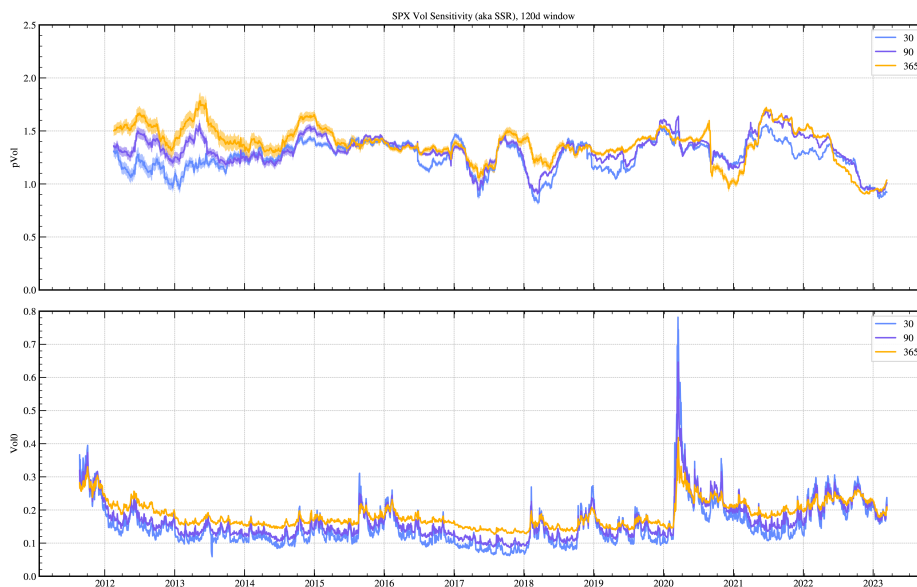


Figure 36: Time series of SSR according to Vola Dynamics. On average $\mathcal{R} \approx 1.5$.

The average term structure of the SSR

Here is another figure from a paper by Bourgey et al. [\[2\]](#)

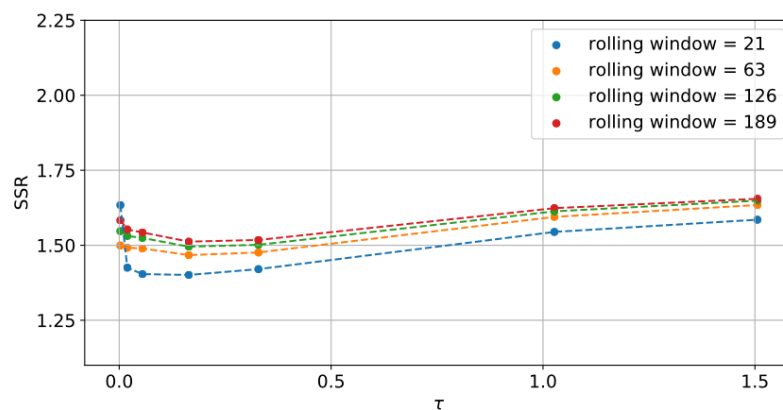


Figure 2: Average realized SSR for the SPX Index. We display the term structure of the ATMF SSR, averaged over the entire time series shown in Figure 1 (i.e., from 2008 to 2015).

Figure 37: Average term structure of the SSR from 2008 to 2015.

Skew-stickiness summary

- The fact that the skew-stickiness ratio (SSR) is approximately independent of expiry on each day is a striking empirical fact.

- Values of the SSR close to 1 at first seem surprising, and are certainly incompatible with popular stochastic volatility models.
 - However, there are stochastic volatility models that fit SPX smiles, VIX smiles and match the SSR!

The variance swap curve

Assuming the underlying diffuses, the fair value $\mathcal{V}_t(T)$ of a variance swap with maturity T may be computed as follows:

$$\mathcal{V}_t(T) = \int_t^T \xi_t(u) du = \int_0^\infty \min[P(K, T), C(K, T)] \frac{dK}{K^2}$$

where $C(K, T)$ and $P(K, T)$ denote the prices of calls and puts with strike K and expiration T respectively. For details, see for example [The Volatility Surface]^[5]. We will visit this again later in the lecture series.

Thus, with this continuity assumption, we may compute the fair value of a variance swap from an infinite strip of call and put option prices (the so-called log-strip).

- For a given maturity, the variance swap level depends on exactly how we interpolate and extrapolate option prices.
 - We use the *arbitrage-free SVI parameterization* that I will also explain later in the lecture series.
- We will analyze variance swap estimates from January 01, 2010 to February 28, 2023.

```
In [76]: load("vsFixedExpiry20230820.rData")

vss <- vs.fixed.expiry[(vs.fixed.expiry[,1]>"201000101"),]

mn <- function(vec){mean(as.numeric(vec),na.rm=T)}
sdd <- function(vec){sd(as.numeric(vec),na.rm=T)}

expiries <- (1:40)/20

vs.mean <- apply(vss[,-1],2,mn)
vs.sd <- apply(vss[,-1],2,sdd)
```

The average shape of the variance swap curve

```
In [77]: plot(expiries,sqrt(vs.mean),type="b",lwd=2,col="royalblue",pch=20,
             xlab=expression(paste("Maturity ",tau)),ylab="Variance swap quote")
```

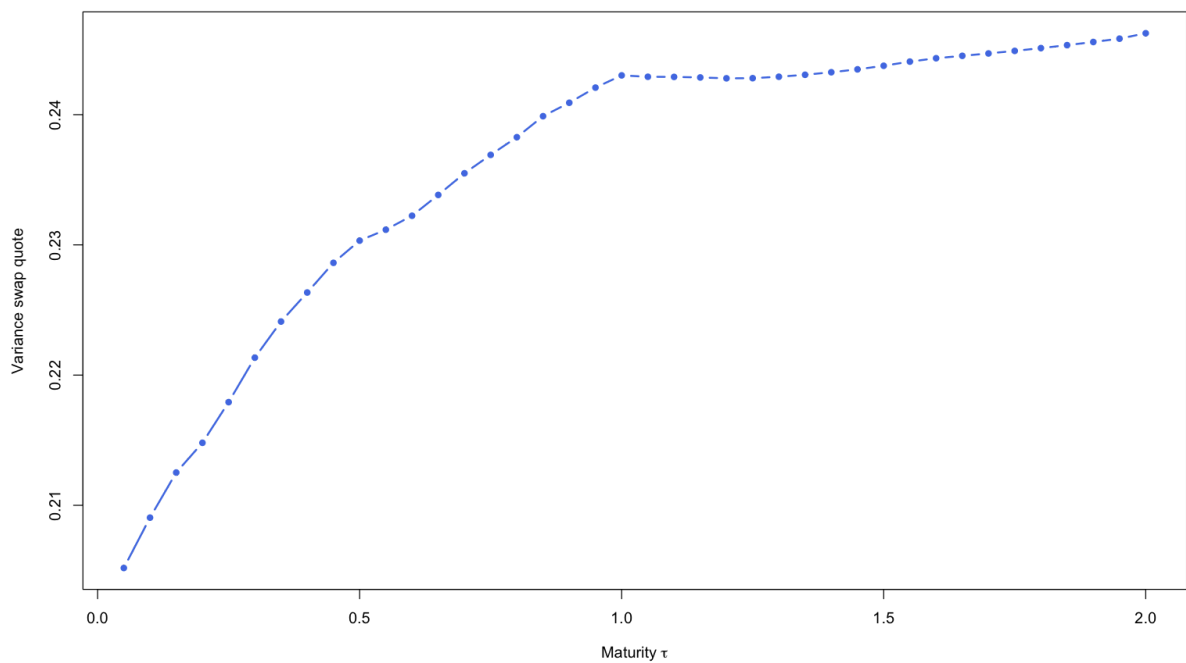


Figure 38: Average shape of variance swap curve from 01-Jan-2010 to 28-Feb-2023.

The variance swap envelope

We define the *envelope* to be the graph of the standard deviation of variance swap quotes vs time to maturity.

```
In [78]: plot(log(expiries), log(vs.sd), col="red2", pch=20, cex=2,
             xlab=expression(paste("log(Maturity)")), ylab="log(Standard deviation of
```

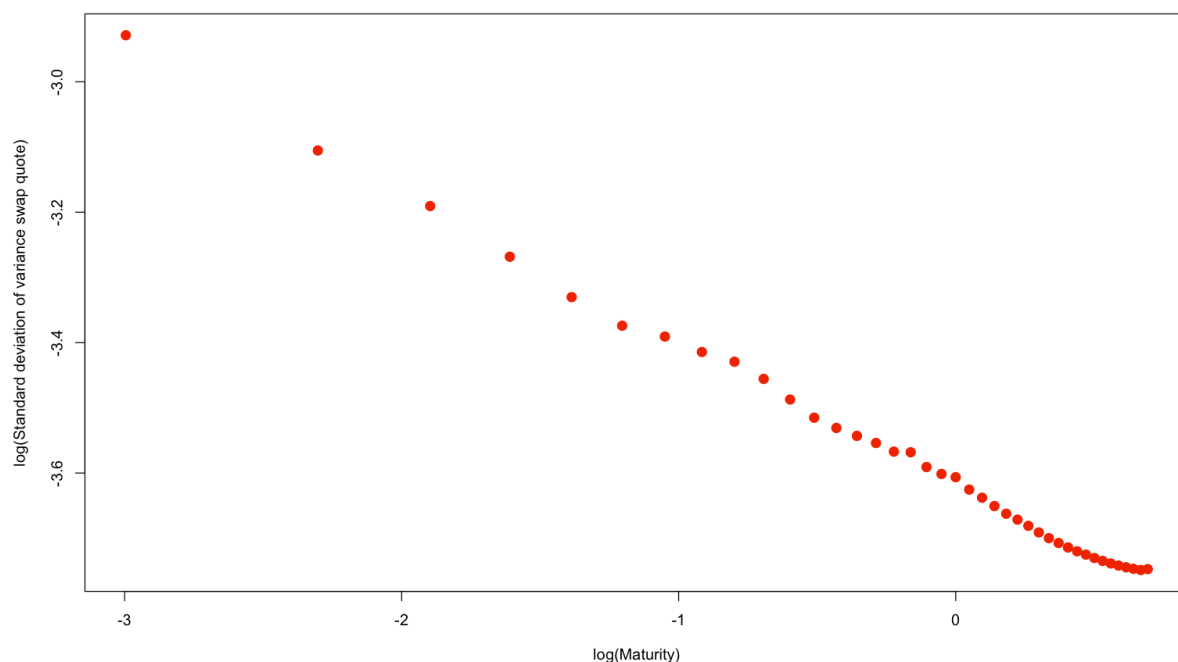


Figure 39: Log-log plot of volatility envelope.

Find the slope of the line

```
In [79]: include <- c(1:2,4:20)
fit.lm <- lm(log(vs.sd[include])~log(expiries[include]))

a <- fit.lm$coef[1]
b <- fit.lm$coef[2]

summary(fit.lm)
```

Call:

```
lm(formula = log(vs.sd[include]) ~ log(expiries[include]))
```

Residuals:

Min	1Q	Median	3Q	Max
-0.0236365	-0.0070592	0.0003322	0.0071989	0.0225994

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-3.618645	0.003974	-910.61	<2e-16 ***
log(expiries[include])	-0.222754	0.003507	-63.52	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.01187 on 17 degrees of freedom

Multiple R-squared: 0.9958, Adjusted R-squared: 0.9956

F-statistic: 4035 on 1 and 17 DF, p-value: < 2.2e-16

Variance swap envelope with power-law fit

```
In [80]: plot(expiries,vs.sd,col="red2",pch=20,xlab=expression(paste("Maturity ",tau))
curve(exp(a+b*log(x)),from=0,to=3,col="blue",add=T,n=1000,lwd=2)
```

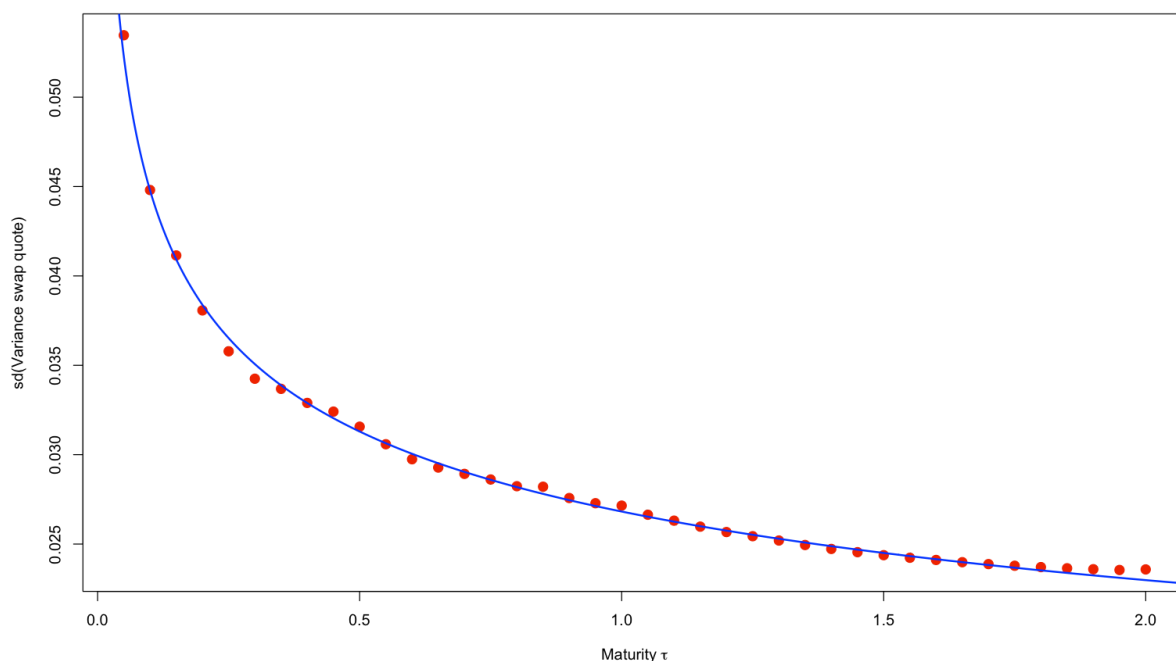


Figure 40: Variance swap envelope from 01-Jan-2010 to 28-Feb-2023. The blue line corresponds to the fit $\tau^{-0.23}$.

Volatility surface summary

- The term structure of ATM volatility skew seems behave as a power-law $\sim \tau^{-0.4}$ for $\tau < 1$.
- The term structure of the volatility envelope seems to be power-law $\sim \tau^{-0.25}$ for $\tau < 1$.
- The Skew-stickiness ratio is approximately independent of time to expiry but its value variss over time between 1 and 1.5.

Before we start modeling

Stochastic volatility models are well-motivated: returns normalized by integrated variance are close to $N(0, 1)$.

Conventional Markovian stochastic volatility models are nevertheless *normative*.

- We write down underlying dynamics as if to say "suppose the underlying stochastic drivers were to satisfy the following...".

- The dynamics are invariably Markovian. The world we have just been looking at is not.
- The state space is typically very small.
- Conventional stochastic volatility models are thus Markovian approximations to a non-Markovian reality.

Conventional models of volatility are engineering models, not physics models.

References

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